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Revision History

Rev.	Descriptions	Date	By
X1	*Draft version*	*04/15/2022*	*Jonson Chen*
A	*Updated according to the review comments.*	*6/30/2022*	*Jonson Chen*
B	*Fixed supply voltage issue on buffer U34, U46, U47, U53, U49.*	*9/27/2022*	*Jonson Chen*
B1	*Changed R80, R81 value to 0 ohm on USB DM/DP pins.*	*10/21/2022*	*Jonson Chen*
B2	*Changed VDD_MCU to VDD_BRD in MCULink VCOM* *Changed TP29/30/31/32/61/62 type to TEST POINT WHITE.* *Changed R274 from DNP to populate.* *Added a jumper on Y7 to disable the output.* *Added SAI signals on M.2 connector.* *Changed external flash from MT35X to W25Q64JW* *Changed PMOS to load switch for VDD_SD control* *Removed redundant resistors, added I2C address table.*	*01/13/2023*	*Jonson Chen*
BX1	*Changed U51 from NTB0101GM,115 to TXB0101DCKR* *Chnaged J7 from STX-35331-4N-TR to SJ2-3574A-SMT-TR* *Changed J20 from 08010S32811-29LF to PPPC142LFBN-RC* *Changed U30 from TJA1057GT/3 to TJA1057BT/0Z* *Changed default Jumper for JP24* *Changed U19 from TJA1057BT/0Z to LTC2051IMS8#PBF* *Changed JP35 from 1x3 to 1x2 header* *Made JP48 DNP*	*11/21/2023*	*Sathiyasri*
B3	*A085 Release*	*12/11/2023*	*Sathiyasri*
B4	*Changed block diagram* *Changed C46 on VDD_CORE from 1uF to 4.7uF* *Changed C53 on VDD_SYS from 0.1uF to 2.2uF*	*1/26/2024*	*Jonson Chen*

MCX-N5XX-EVK
55277-B4

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Designer: Jonson Chen	Drawing Title: MCX-N5XX-EVK	
Drawn by: Jonson Chen	Page Title: Titel,TOC, Rev History	
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Jumper Setting

REF DES	*JUMPER(DEFAULT)*	*PAGE NAME*
JP13,JP17	1-2	04 MCU_PORT_0_1_2_3
JP21	2-3	04 MCU_PORT_0_1_2_3
JP14,JP26,JP27,JP30,JP36, JP37,JP39,JP40,JP45,JP47	1-2	06 MCU_POWER
JP18	2-3	06 MCU_POWER
JP31,JP35	OPEN	06 MCU_POWER
J26,JP43,JP44,JP46	1-2	07 PMIC
JP41	OPEN	07 PMIC
JP24	1-2	08 MCU_LINK_USB
JP5,JP6,JP19	OPEN	08 MCU_LINK_USB
JP12	1-2	09 MCU_LINK_DEBUG
JP9,JP22	OPEN	09 MCU_LINK_DEBUG
JP32,JP33,JP38	1-2	10 MCU_LINK_CURRENT
JP25	1-2	11 MCU_LINK_VCOM_SIO
JP51	1-2	12 USB
JP1,JP3,JP7,JP8,JP10,JP11, JP16,JP20	1-2	14 AUDIO
R897,R898,R899,R900,R901, R902,R903,R904	1-2	15 MEMORY
JP4	1-2	16 ETHERNET
JP50	OPEN	16 ETHERNET
JP34	1-2	17 SENSOR_CAN
JP28	OPEN	18 HEADERS

DNP List

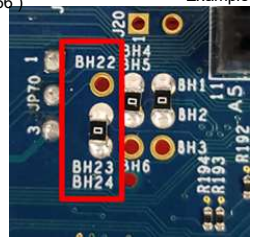
REF DES	*ASSY_OPT*	*PAGE NAME*
R11,R12,R13,R14,R19,R20,R23, R816,R822,R823,R824,R825, R826,R830,R835,R836,R860, R861,R862,R863,R864,R865, R882,R885,R925	DNP	04 MCU_PORT_0_1_2_3
R76,R77,R303,R307,R869,R870, R874	DNP	05 MCU_PORT_4_5_A
JP15	DNP	06 MCU_POWER
JP48,R86,R87,R89,R90,R362	DNP	07 PMIC
C89,C90,J13,J15,R107,R115, R116	DNP	08 MCU_LINK_USB
D16,D17,D18,R123,R125	DNP	09 MCU_LINK_DEBUG
R145,R158	DNP	10 MCU_LINK_CURRENT
R179,R180,R186,R187,R909, R975,R976,R977,R978	DNP	12 USB
C144,C145,R194,R196,R197,R198	DNP	13 CLOCK_RESET_HMI
R937	DNP	14 AUDIO
C400,C401,J9,R469,R474,R475, R479,R483,R487,R929,R930, R931,R932,R934,R954,R956, R957,R958,R960,R961,R963, R964,R966,R967	DNP	15 MEMORY
C186,C193,C194,C195,C196, R276,R277,R278,R279,R280, R281,R282,R283	DNP	16 ETHERNET
C202,C209,C411,D23,J17,J18, J29,R288,R289,R290,R291,R895, R896,R969,R970,R971,U39,U50	DNP	17 SENSOR_CAN
C213,C214,C215,C216,J23,R311, R313,R948,R949,R951	DNP	18 HEADERS
JP2	DNP	19 M.2_CONNECTOR

CM short resistor (DNP on N5 EVK)

SHORTH Instruccion need to be added at ASY drawing.
Expected resistor :RC0603FR-070RL (470-80866)

Set shorth manually between BH1, BH2
Set shorth manually between BH4, BH5

Example



Device	Part No.	Primary Address (Default)	Primary Address w/ Read bit (Default)	Secondary Address
PMIC	PCA9420BSAZ	0x61	0xC2	0x69
Audio Codec	DA7212	0x1A	0x34	-
Accel/Mag	FXLS8964AF	0x19	0x32	0x18
I3C	P3T1755DP	0x48	0x90	0x49 - 4F
*Addresses are without the R/W bit unless otherwise noted.				

1. Unless Otherwise Specified:

   All resistors are in ohms, 1/16 Watt,0402
   All capacitors are in uF,0402
   All voltages are DC
   All polarized capacitors are aluminum electrolytic

2. Interrupted lines coded with the same letter or letter combinations are electrically connected.

Text Color:
Blue = CAD notes; Green = general notes;

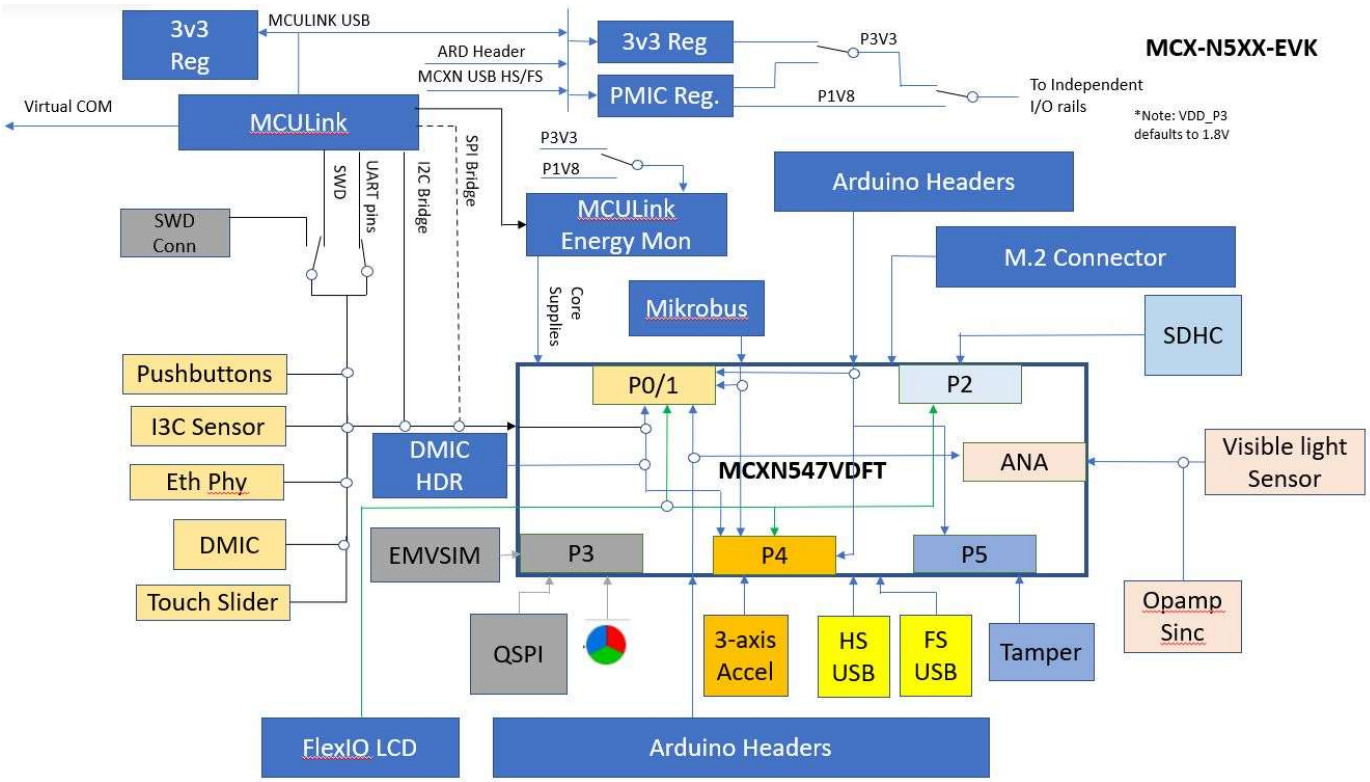
3. Device type number is for reference only. The number varies with the manufacturer.

4. Special signal usage:
   _B Denotes - Active-Low Signal
   <> or [] Denotes - Vectored Signals

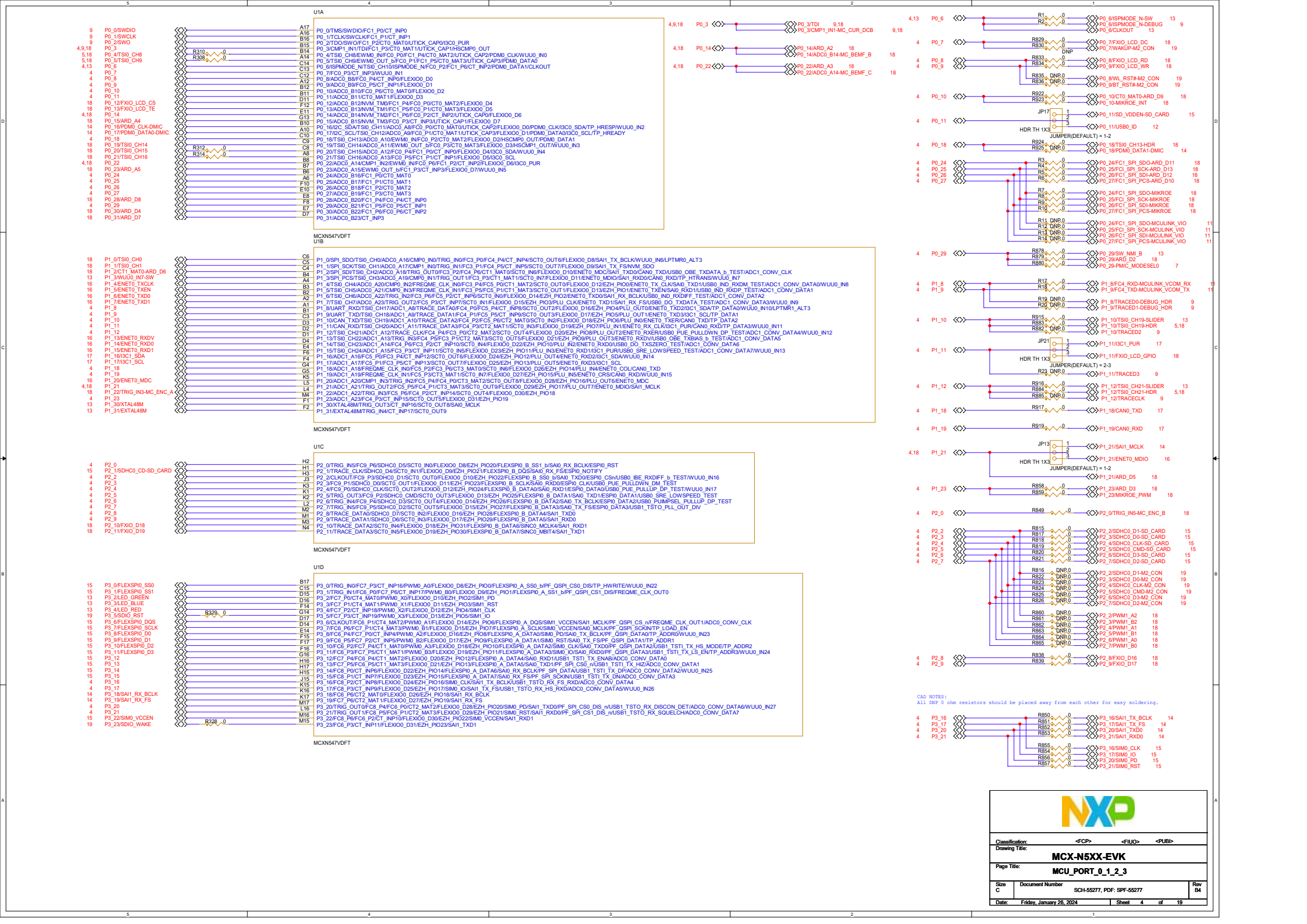
5. Interpret diagram in accordance with American National Standards Institute specifications, current revision, with the exception of logic block symbology.

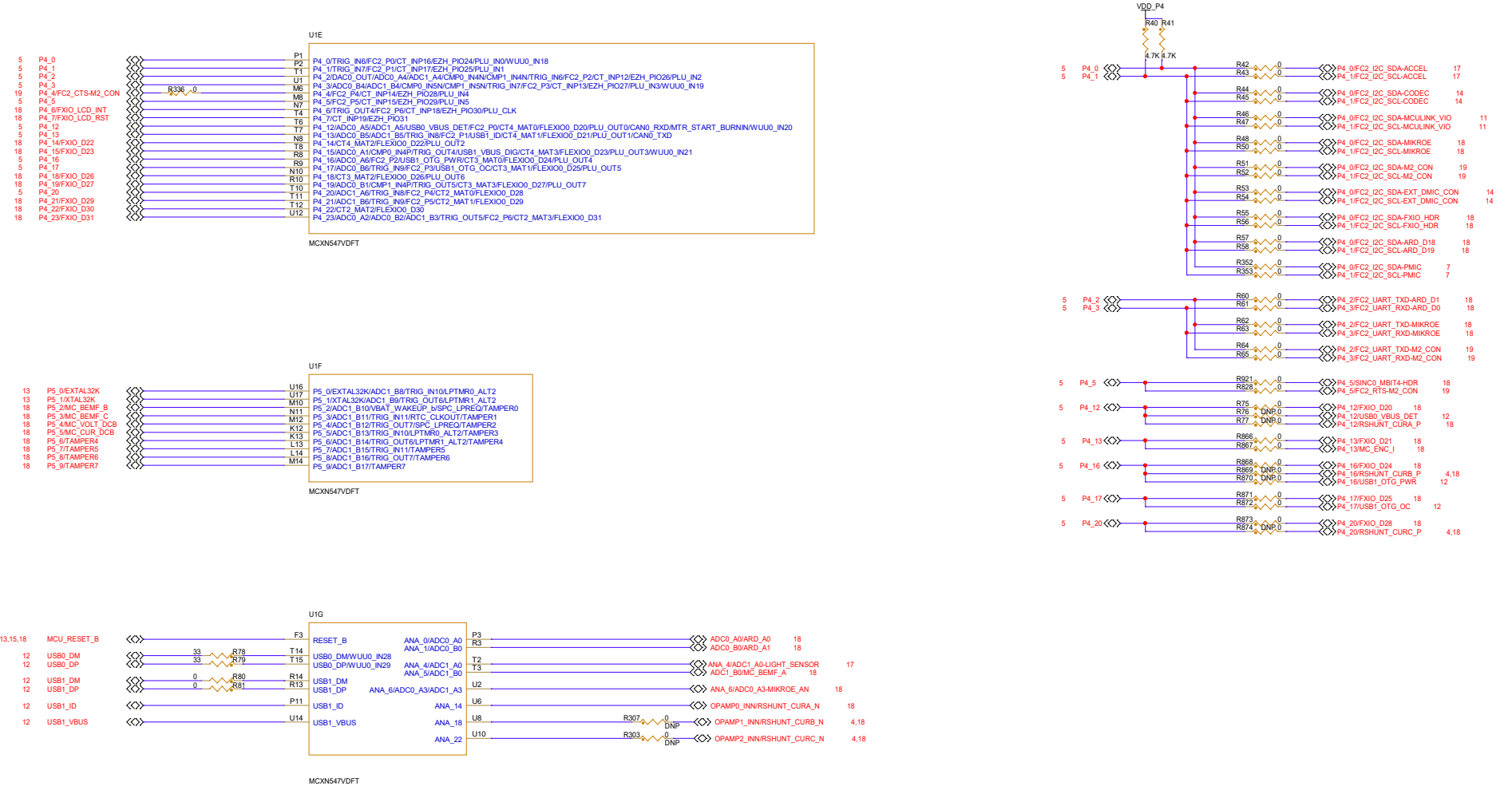
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Drawing Title: MCX-N5XX-EVK
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BLOCK DIAGRAM



Classification:	<FCP>	<FIUO>	<PUBI>
Drawing Title:	MCX-N5XX-EVK		
Page Title:	BLOCK DIAGRAM		
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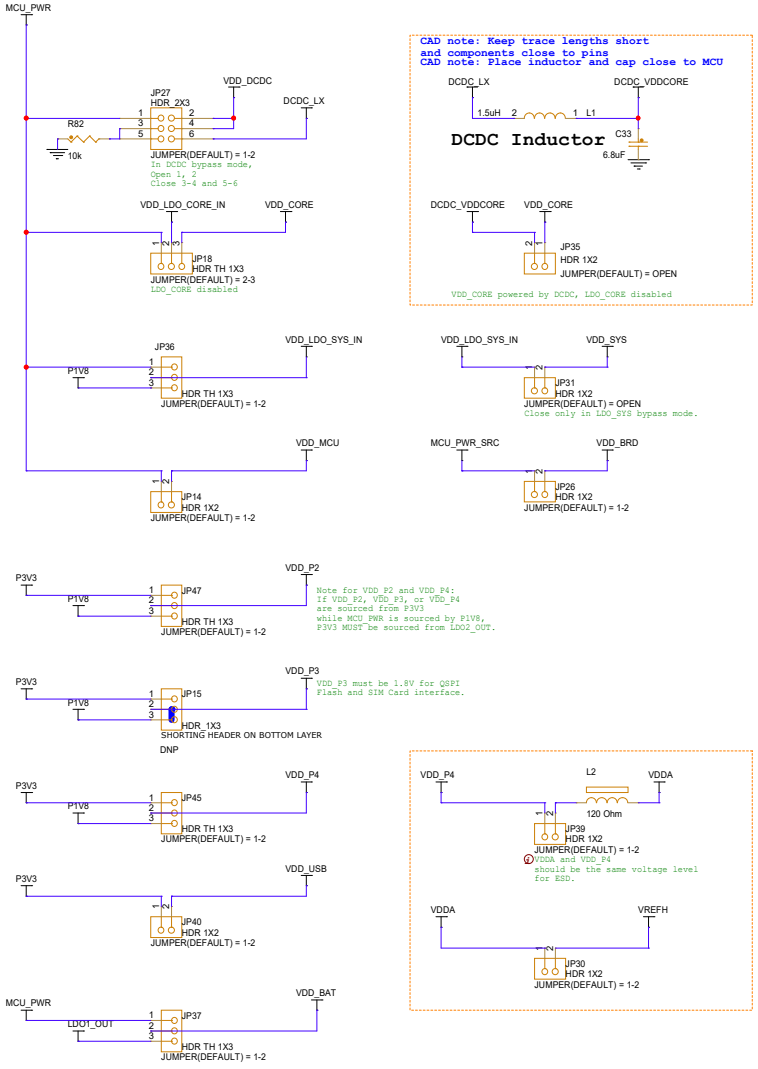

U1A
P0_0/TMS/SWDIO/FC1_P0/CT_INP0
P0_1/TCLK/SWCLK/FC1_P1/CT_INP1
P0_2/TDO/SWO/FC1_P2/CT0_MAT0/UTICK_CAP0/I3C0_PUR
P0_3/CMP1_IN1/TDI/FC1_P3/CT0_MAT1/UTICK_CAP1/HSCMP0_OUT
P0_4/TSI0_CH8/EWM0_IN/FC0_P0/FC1_P4/CT0_MAT2/UTICK_CAP2/PDM0_CLK/WUU0_IN0
P0_5/TSI0_CH9/EWM0_OUT_b/FC0_P1/FC1_P5/CT0_MAT3/UTICK_CAP3/PDM0_DATA0
P0_6/ISPMODE_N/TSI0_CH10/ISPMODE_N/FC0_P2/FC1_P6/CT_INP2/PDM0_DATA1/CLKOUT
P0_7/FC0_P3/CT_INP3/WUU0_IN1
P0_8/ADC0_B8/FC0_P4/CT_INP0/FLEXIO0_D0
P0_9/ADC0_B9/FC0_P5/CT_INP1/FLEXIO0_D1
P0_10/ADC0_B10/FC0_P6/CT0_MAT0/FLEXIO0_D2
P0_11/ADC0_B11/CT0_MAT1/FLEXIO0_D3
P0_12/FXIO_LCD_CS
P0_13/FXIO_LCD_TE
P0_14
P0_15/ARD_A4
P0_16/PDM0_CLK-DMIC
P0_17/PDM0_DATA0-DMIC
P0_18
P0_19/TSI0_CH14
P0_20/TSI0_CH15
P0_21/TSI0_CH16
P0_22
P0_23/ARD_A5
P0_24
P0_25
P0_26
P0_27
P0_28/ARD_D8
P0_29
P0_30/ARD_D4
P0_31/ARD_D7
R310 0
R308 0
R312 0
R314 0
MCXN547VFDT
U1B
P1_0/SPI_SDO/TSI0_CH0/ADC0_A16/CMP0_IN0/TRIG_IN0/FC3_P0/FC4_P4/CT_INP4/SCT0_OUT6/FLEXIO0_D8/SAI1_TX_BCLK/WUU0_IN6/LPTMR0_ALT3
P1_1/SPI_SCK/TSI0_CH1/ADC0_A17/CMP1_IN0/TRIG_IN1/FC3_P1/FC4_P5/CT_INP5/SCT0_OUT7/FLEXIO0_D9/SAI1_TX_FS/NVM_SDO
P1_2/SPI_SDI/TSI0_CH2/ADC0_A18/TRIG_OUT0/FC3_P2/FC4_P6/CT1_MAT0/SCT0_IN6/FLEXIO0_D10/ENET0_MDC/SAI1_TXD0/CAN0_TXD/USB0_OBE_TXDATA_b_TEST/ADC1_CONV_CLK
P1_3/SPI_PCS/TSI0_CH3/ADC0_A19/CMP0_IN1/TRIG_OUT1/FC3_P3/CT1_MAT1/SCT0_IN7/FLEXIO0_D11/ENET0_MDIO/SAI1_RXD0/CAN0_RXD/TP_HTRANS/WUU0_IN7
P1_4/ENET0_TXCLK/ADC0_A20/CMP0_IN2/FREQME_CLK_IN0/FC3_P4/FC5_P0/CT1_MAT2/SCT0_OUT0/FLEXIO0_D12/EZH_PIO0/ENET0_TX_CLK/SAI0_TXD1/USB0_IND_RXDM_TEST/ADC1_CONV_DATA0/WUU0_IN8
P1_5/TSI0_CH5/ADC0_A21/CMP0_IN3/FREQME_CLK_IN1/FC3_P5/FC5_P1/CT1_MAT3/SCT0_OUT1/FLEXIO0_D13/EZH_PIO1/ENET0_TXEN/SAI0_RXD1/USB0_IND_RXDP_TEST/ADC1_CONV_DATA1
P1_6/TSI0_CH6/ADC0_A22/TRIG_IN2/FC3_P6/FC5_P2/CT_INP6/SCT0_IN0/FLEXIO0_D14/EZH_PIO2/ENET0_TXD0/SAI1_RX_BCLK/USB0_IND_RXDIFF_TEST/ADC1_CONV_DATA2
P1_7/TSI0_CH7/ADC0_A23/TRIG_OUT2/FC5_P3/CT_INP7/SCT0_IN1/FLEXIO0_D15/EZH_PIO3/PLU_CLK/ENET0_TXD1/SAI1_RX_FS/USB0_DO_TXDATA_TEST/ADC1_CONV_DATA3/WUU0_IN9
P1_8/UART_TXD/TSI0_CH17/ADC1_A8/TRACE_DATA0/FC4_P0/FC5_P4/CT_INP8/SCT0_OUT2/FLEXIO0_D16/EZH_PIO4/PLU_OUT0/ENET0_TXD2/I3C1_SDA/TP_DATA0/WUU0_IN10/LPTMR1_ALT3
P1_9/UART_TXD/TSI0_CH18/ADC1_A9/TRACE_DATA1/FC4_P1/FC5_P5/CT_INP9/SCT0_OUT3/FLEXIO0_D17/EZH_PIO5/PLU_OUT1/ENET0_TXD3/I3C1_SCL/TP_DATA1
P1_10/CAN_TXD/TSI0_CH19/ADC1_A10/TRACE_DATA2/FC4_P2/FC5_P6/CT2_MAT0/SCT0_IN2/FLEXIO0_D18/EZH_PIO6/PLU_IN0/ENET0_TXER/CAN0_TXD/TP_DATA2
P1_11/CAN_RXD/TSI0_CH20/ADC1_A11/TRACE_DATA3/FC4_P3/CT2_MAT1/SCT0_IN3/FLEXIO0_D19/EZH_PIO7/PLU_IN1/ENET0_RX_CLK/I3C1_PUR/CAN0_RXD/TP_DATA3/WUU0_IN11
P1_12/TSI0_CH4/ADC1_A12/TRACE_CLK/FC4_P4/FC3_P0/CT2_MAT2/SCT0_OUT4/FLEXIO0_D20/EZH_PIO8/PLU_IN2/ENET0_RXER/USB0_PUE_PULLDWN_DP_TEST/ADC1_CONV_DATA4/WUU0_IN12
P1_13/TSI0_CH22/ADC1_A13/TRIG_IN3/FC4_P5/FC3_P1/CT2_MAT3/SCT0_OUT5/FLEXIO0_D21/EZH_PIO9/PLU_OUT3/ENET0_RXDV/USB0_OBE_TXBIAS_b_TEST/ADC1_CONV_DATA5
P1_14/TSI0_CH23/ADC1_A14/FC4_P6/FC3_P2/CT_INP10/SCT0_IN4/FLEXIO0_D22/EZH_PIO10/PLU_IN2/ENET0_RXD0/USB0_DO_TXSZERO_TEST/ADC1_CONV_DATA6
P1_15/TSI0_CH24/ADC1_A15/FC3_P3/CT_INP11/SCT0_IN5/FLEXIO0_D23/EZH_PIO11/PLU_IN3/ENET0_RXD1/I3C1_PUR/USB0_SRE_LOWSPEED_TEST/ADC1_CONV_DATA7/WUU0_IN13
P1_16/ADC1_A16/FC5_P0/FC3_P4/CT_INP12/SCT0_OUT6/FLEXIO0_D24/EZH_PIO12/PLU_OUT4/ENET0_RXD2/I3C1_SDA/WUU0_IN14
P1_17/ADC1_A17/FC5_P1/FC3_P5/CT_INP13/SCT0_OUT7/FLEXIO0_D25/EZH_PIO13/PLU_OUT5/ENET0_RXD3/I3C1_SCL
P1_18/ADC1_A18/FREQME_CLK_IN0/FC5_P2/FC3_P6/CT3_MAT0/SCT0_IN6/FLEXIO0_D26/EZH_PIO14/PLU_IN4/ENET0_COL/CAN0_TXD
P1_19/ADC1_A19/FREQME_CLK_IN1/FC5_P3/CT3_MAT1/SCT0_IN7/FLEXIO0_D27/EZH_PIO15/PLU_IN5/ENET0_CRS/CAN0_RXD/WUU0_IN15
P1_20/ADC1_A20/CMP1_IN3/TRIG_IN2/FC5_P4/FC4_P0/CT3_MAT2/SCT0_OUT8/FLEXIO0_D28/EZH_PIO16/PLU_OUT6/ENET0_MDC
P1_21/ADC1_A21/TRIG_OUT2/FC5_P5/FC4_P1/CT3_MAT3/SCT0_OUT9/FLEXIO0_D29/EZH_PIO17/PLU_OUT7/ENET0_MDIO/SAI1_MCLK
P1_22/ADC1_A22/TRIG_IN3/FC5_P6/FC4_P2/CT_INP14/SCT0_OUT4/FLEXIO0_D30/EZH_PIO18
P1_23/ADC1_A23/FC4_P3/CT_INP15/SCT0_OUT5/FLEXIO0_D31/EZH_PIO19
P1_30/XTAL48M/TRIG_OUT3/CT_INP16/SCT0_OUT8/SAI0_MCLK
P1_31/EXTAL48M/TRIG_IN4/CT_INP17/SCT0_OUT9
MCXN547VFDT
U1C
P2_0/TRIG_IN5/FC9_P6/SDHC0_D5/SCT0_IN0/FLEXIO0_D8/EZH_PIO20/FLEXSPI0_B_SS1_b/SAI0_RX_BCLK/ESPI0_RST
P2_1/TRACE_CLK/SDHC0_D4/SCT0_IN1/FLEXIO0_D9/EZH_PIO21/FLEXSPI0_B_DQS/SAI0_RX_FS/ESPI0_NOTIFY
P2_2/CLKOUT/FC9_P3/SDHC0_D1/SCT0_OUT0/FLEXIO0_D10/EZH_PIO22/FLEXSPI0_B_SS0_b/SAI0_TXD0/ESPI0_CSn/USB0_IBE_RXDIFF_b_TEST/WUU0_IN16
P2_3/FC9_P1/SDHC0_D0/SCT0_OUT1/FLEXIO0_D11/EZH_PIO23/FLEXSPI0_B_SCLK/SAI0_RXD0/ESPI0_CLK/USB0_PUE_PULLDWN_DM_TEST
P2_4/FC9_P0/SDHC0_CLK/SCT0_OUT2/FLEXIO0_D12/EZH_PIO24/FLEXSPI0_B_DATA0/SAI0_RXD1/ESPI0_DATA0/USB0_PUE_PULLUP_DP_TEST/WUU0_IN17
P2_5/TRIG_OUT3/FC9_P2/SDHC0_CMD/SCT0_OUT3/FLEXIO0_D13/EZH_PIO25/FLEXSPI0_B_DATA1/SAI0_TXD1/ESPI0_DATA1/USB0_SRE_LOWSPEED_TEST
P2_6/TRIG_IN4/FC9_P4/SDHC0_D3/SCT0_OUT4/FLEXIO0_D14/EZH_PIO26/FLEXSPI0_B_DATA2/SAI0_TX_BCLK/ESPI0_DATA2/USB0_PUIMPSEL_PULLUP_DP_TEST
P2_7/TRIG_IN5/FC9_P5/SDHC0_D2/SCT0_OUT5/FLEXIO0_D15/EZH_PIO27/FLEXSPI0_B_DATA3/SAI0_TX_FS/ESPI0_DATA3/USB1_TSTO_PLL_OUT_DIV
P2_8/TRACE_DATA0/SDHC0_D7/SCT0_IN2/FLEXIO0_D16/EZH_PIO28/FLEXSPI0_B_DATA4/SAI1_TXD0
P2_9/TRACE_DATA1/SDHC0_D6/SCT0_IN3/FLEXIO0_D17/EZH_PIO29/FLEXSPI0_B_DATA5/SAI1_RXD0
P2_10/TRACE_DATA2/SCT0_IN4/FLEXIO0_D18/EZH_PIO31/FLEXSPI0_B_DATA6/SINC0_MCLK4/SAI1_RXD1
P2_11/TRACE_DATA3/SCT0_IN5/FLEXIO0_D19/EZH_PIO30/FLEXSPI0_B_DATA7/SINC0_MBIT4/SAI1_TXD1
MCXN547VFDT
U1D
P3_0/TRIG_IN0/FC7_P3/CT_INP16/PWM0_A0/FLEXIO0_D8/EZH_PIO0/FLEXSPI0_A_SS0_b/PF_QSPI_CS0_DIS/TP_HWRITE/WUU0_IN22
P3_1/TRIG_IN1/FC6_P0/FC7_P6/CT_INP17/PWM0_B0/FLEXIO0_D9/EZH_PIO1/FLEXSPI0_A_SS1_b/PF_QSPI_CS1_DIS/FREQME_CLK_OUT0
P3_2/FC7_P0/CT4_MAT0/PWM0_X0/FLEXIO0_D10/EZH_PIO2/SIM1_PD
P3_3/FC7_P1/CT4_MAT1/PWM0_X1/FLEXIO0_D11/EZH_PIO3/SIM1_RST
P3_4/FC7_P2/CT_INP18/PWM0_X2/FLEXIO0_D12/EZH_PIO4/SIM1_CLK
P3_5/FC7_P3/CT_INP19/PWM0_X3/FLEXIO0_D13/EZH_PIO5/SIM1_IO
P3_6/CLKOUT/FC6_P1/CT4_MAT2/PWM0_A1/FLEXIO0_D14/EZH_PIO6/FLEXSPI0_A_DQS/SIM1_VCCEN/SAI1_MCLK/PF_QSPI_CS_n/FREQME_CLK_OUT1/ADC0_CONV_CLK
P3_7/FC6_P6/FC7_P1/CT4_MAT3/PWM0_B1/FLEXIO0_D15/EZH_PIO7/FLEXSPI0_A_SCLK/SIM0_VCCEN/SAI0_MCLK/PF_QSPI_SCKIN/TP_LOAD_EN
P3_8/FC6_P4/FC7_P0/CT_INP4/PWM0_A2/FLEXIO0_D16/EZH_PIO8/FLEXSPI0_A_DATA0/SIM0_PD/SAI0_TX_BCLK/PF_QSPI_DATA0/TP_ADDR0/WUU0_IN23
P3_9/FC6_P5/FC7_P2/CT_INP5/PWM0_B2/FLEXIO0_D17/EZH_PIO9/FLEXSPI0_A_DATA1/SIM0_RST/SAI0_TX_FS/PF_QSPI_DATA1/TP_ADDR1
P3_10/FC8_P2/FC7_P4/CT1_MAT0/PWM0_A3/FLEXIO0_D18/EZH_PIO10/FLEXSPI0_A_DATA2/SIM0_CLK/SAI0_TXD0/PF_QSPI_DATA2/USB1_TSTI_TX_HS_MODE/TP_ADDR2
P3_11/FC6_P3/FC7_P5/CT1_MAT1/PWM0_B3/FLEXIO0_D19/EZH_PIO11/FLEXSPI0_A_DATA3/SIM0_IO/SAI0_RXD0/PF_QSPI_DATA3/USB1_TSTI_TX_LS_EN/TP_ADDR3/WUU0_IN24
P3_12/FC7_P6/CT1_MAT2/PWM0_X0/FLEXIO0_D20/EZH_PIO12/FLEXSPI0_A_DATA4/SAI0_RXD1/USB1_TSTI_TX_ENAB/ADC0_CONV_DATA0
P3_13/FC7_P5/FC6_P5/CT1_MAT3/FLEXIO0_D21/EZH_PIO13/FLEXSPI0_A_DATA5/SAI0_TXD1/PF_SPI_CS0_n/USB1_TSTI_TX_HIZ/ADC0_CONV_DATA1
P3_14/FC8_P0/CT_INP6/FLEXIO0_D22/EZH_PIO14/FLEXSPI0_A_DATA6/SAI0_RX_BCLK/PF_SPI_DATA/USB1_TSTI_TX_DP/ADC0_CONV_DATA2/WUU0_IN25
P3_15/FC8_P1/CT_INP7/FLEXIO0_D23/EZH_PIO15/FLEXSPI0_A_DATA7/SAI0_RX_FS/PF_SPI_SCKIN/USB1_TSTI_TX_DN/ADC0_CONV_DATA3
P3_16/FC8_P2/CT_INP8/FLEXIO0_D24/EZH_PIO16/SIM0_CLK/SAI1_TX_BCLK/USB1_TSTO_RX_FS_RXD/ADC0_CONV_DATA4
P3_17/FC8_P3/CT_INP9/FLEXIO0_D25/EZH_PIO17/SIM0_IO/SAI1_TX_FS/USB1_TSTO_RX_HS_RXD/ADC0_CONV_DATA5/WUU0_IN26
P3_18/FC6_P6/CT2_MAT0/FLEXIO0_D26/EZH_PIO18/SAI1_RX_BCLK
P3_19/FC7_P6/CT2_MAT1/FLEXIO0_D27/EZH_PIO19/SAI1_RX_FS
P3_20/TRIG_OUT0/FC8_P4/FC6_P0/CT2_MAT2/FLEXIO0_D28/EZH_PIO20/SIM0_PD/SAI1_TXD0/PF_SPI_CS0_DIS_n/USB1_TSTO_RX_DISCON_DET/ADC0_CONV_DATA6/WUU0_IN27
P3_21/TRIG_OUT1/FC8_P5/FC6_P1/CT2_MAT3/FLEXIO0_D29/EZH_PIO21/SIM0_RST/SAI1_RXD0/PF_SPI_CS1_DIS_n/USB1_TSTO_RX_SQUELCH/ADC0_CONV_DATA7
P3_22/FC8_P6/FC6_P2/CT_INP10/FLEXIO0_D30/EZH_PIO22/SIM0_VCCEN/SAI1_RXD1
P3_23/FC8_P3/CT_INP11/FLEXIO0_D31/EZH_PIO23/SAI1_TXD1
R329 0
R328 0
MCXN547VFDT
P0_3/TDI 9,18
P0_3/CMP1_IN1-MC_CUR_DCB 9,18
P0_14/ARD_A2 18
P0_14/ADC0_B14-MC_BEMF_B 18
P0_22/ARD_A3 18
P0_22/ADC0_A14-MC_BEMF_C 18
R1 0
R2 0
R829 0
R830 0
DNP
R833 0
R834 0
R835 DNP 0
R836 DNP 0
R922 0
R923 0
JP17 1 2 3
HDR TH 1X3
JUMPER(DEFAULT) = 1-2
R924 0
R925 DNP 0
R3 0
R4 0
R5 0
R6 0
R7 0
R8 0
R9 0
R10 0
R11 DNP 0
R12 DNP 0
R13 DNP 0
R14 DNP 0
R878 0
R879 0
R880 0
R17 0
R18 0
R19 DNP 0
R20 DNP 0
R915 0
R883 0
R882 DNP 0
JP21 1 2 3
HDR TH 1X3
JUMPER(DEFAULT) = 2-3
R23 DNP 0
R916 0
R884 0
R885 DNP 0
R917 0
R919 0
JP13 1 2 3
HDR TH 1X3
JUMPER(DEFAULT) = 1-2
R858 0
R859 0
R849 0
R815 0
R817 0
R818 0
R819 0
R820 0
R821 0
R816 DNP 0
R822 DNP 0
R823 DNP 0
R824 DNP 0
R825 DNP 0
R826 DNP 0
R860 DNP 0
R861 DNP 0
R862 DNP 0
R863 DNP 0
R864 DNP 0
R865 DNP 0
R838 0
R839 0
R850 0
R851 0
R852 0
R853 0
R855 0
R854 0
R856 0
R857 0
4,13 P0_6
4 P0_7
4 P0_8
4 P0_9
4 P0_10
4 P0_11
4 P0_18
4 P0_24
4 P0_25
4 P0_26
4 P0_27
4 P0_29
4 P1_8
4 P1_9
4 P1_10
4 P1_11
4 P1_12
4 P1_18
4 P1_19
4,18 P1_21
4 P1_23
4 P2_0
4 P2_2
4 P2_3
4 P2_4
4 P2_5
4 P2_6
4 P2_7
4 P2_8
4 P2_9
4 P3_16
4 P3_17
4 P3_20
4 P3_21
P0_6/ISPMODE_N-SW 13
P0_6/ISPMODE_N-DEBUG 9
P0_6/CLKOUT 13
P0_7/FXIO_LCD_DC 18
P0_7/WAKUP-M2_CON 19
P0_8/FXIO_LCD_RD 18
P0_9/FXIO_LCD_WR 18
P0_8/WL_RST#-M2_CON 19
P0_9/BT_RST#-M2_CON 19
P0_10/CT0_MAT0-ARD_D9 18
P0_10-MIKROE_INT 18
P0_11/SD_VDDEN-SD_CARD 15
P0_11/USB0_ID 12
P0_18/TSI0_CH13-HDR 18
P0_18/PDM0_DATA1-DMIC 14
P0_24/FC1_SPI_SDO-ARD_D11 18
P0_25/FC1_SPI_SCK-ARD_D13 18
P0_26/FC1_SPI_SDI-ARD_D12 18
P0_27/FC1_SPI_PCS-ARD_D10 18
P0_24/FC1_SPI_SDO-MIKROE 18
P0_25/FC1_SPI_SCK-MIKROE 18
P0_26/FC1_SPI_SDI-MIKROE 18
P0_27/FC1_SPI_PCS-MIKROE 18
P0_24/FC1_SPI_SDO-MCULINK_VIO 11
P0_25/FC1_SPI_SCK-MCULINK_VIO 11
P0_26/FC1_SPI_SDI-MCULINK_VIO 11
P0_27/FC1_SPI_PCS-MCULINK_VIO 11
P0_29/SW_NMI_B 13
P0_29/ARD_D2 18
P0_29-PMIC_MODESEL0 7
P1_8/FC4_RXD-MCULINK_VCOM_RX 11
P1_9/FC4_TXD-MCULINK_VCOM_TX 11
P1_8/TRACED0-DEBUG_HDR 9
P1_9/TRACED1-DEBUG_HDR 9
P1_10/TSI0_CH19-SLIDER 13
P1_10/TSI0_CH19-HDR 5,18
P1_10/TRACED2 9
P1_11/I3C1_PUR 17
P1_11/FXIO_LCD_GPIO 18
P1_11/TRACED3 9
P1_12/TSI0_CH21-SLIDER 13
P1_12/TSI0_CH21-HDR 5,18
P1_12/TRACECLK 9
P1_18/CAN0_TXD 17
P1_19/CAN0_RXD 17
P1_21/SAI1_MCLK 14
P1_21/ENET0_MDIO 16
P1_21/ARD_D5 18
P1_23/ARD_D3 18
P1_23/MIKROE_PWM 18
P2_0/TRIG_IN5-MC_ENC_B 18
P2_2/SDHC0_D1-SD_CARD 15
P2_3/SDHC0_D0-SD_CARD 15
P2_4/SDHC0_CLK-SD_CARD 15
P2_5/SDHC0_CMD-SD_CARD 15
P2_6/SDHC0_D3-SD_CARD 15
P2_7/SDHC0_D2-SD_CARD 15
P2_2/SDHC0_D1-M2_CON 19
P2_3/SDHC0_D0-M2_CON 19
P2_4/SDHC0_CLK-M2_CON 19
P2_5/SDHC0_CMD-M2_CON 19
P2_6/SDHC0_D3-M2_CON 19
P2_7/SDHC0_D2-M2_CON 19
P2_2/PWM1_A2 18
P2_3/PWM1_B2 18
P2_4/PWM1_A1 18
P2_5/PWM1_B1 18
P2_6/PWM1_A0 18
P2_7/PWM1_B0 18
P2_8/FXIO_D16 18
P2_9/FXIO_D17 18
P3_16/SAI1_TX_BCLK 14
P3_17/SAI1_TX_FS 14
P3_20/SAI1_TXD0 14
P3_21/SAI1_RXD0 14
P3_16/SIM0_CLK 15
P3_17/SIM0_IO 15
P3_20/SIM0_PD 15
P3_21/SIM0_RST 15
CAD NOTES:
All DNP 0 ohm resistors should be placed away from each other for easy soldering.
Classification: <FCP> <FIUO> <PUBI>
Drawing Title: MCX-N5XX-EVK
Page Title: MCU_PORT_0_1_2_3
Size C
Document Number SCH-55277, PDF: SPF-55277
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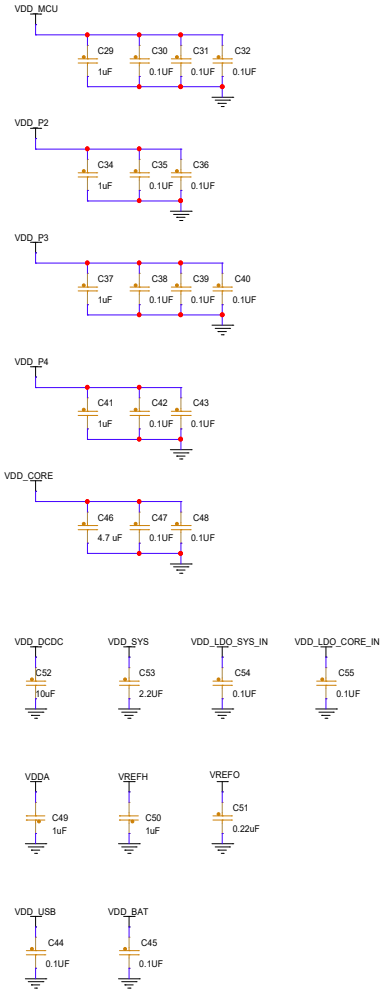

U1E
P4_0/TRIG_IN6/FC2_P0/CT_INP16/EZH_PIO24/PLU_IN0/WUU0_IN18
P4_1/TRIG_IN7/FC2_P1/CT_INP17/EZH_PIO25/PLU_IN1
P4_2/DAC0_OUT/ADC0_A4/ADC1_A4/CMP0_IN4N/CMP1_IN4N/TRIG_IN6/FC2_P2/CT_INP12/EZH_PIO26/PLU_IN2
P4_3/ADC0_B4/ADC1_B4/CMP0_IN5N/CMP1_IN5N/TRIG_IN7/FC2_P3/CT_INP13/EZH_PIO27/PLU_IN3/WUU0_IN19
P4_4/FC2_P4/CT_INP14/EZH_PIO28/PLU_IN4
P4_5/FC2_P5/CT_INP15/EZH_PIO29/PLU_IN5
P4_6/TRIG_OUT4/FC2_P6/CT_INP18/EZH_PIO30/PLU_CLK
P4_7/CT_INP19/EZH_PIO31
P4_12/ADC0_A5/ADC1_A5/USB0_VBUS_DET/FC2_P0/CT4_MAT0/FLEXIO0_D20/PLU_OUT0/CAN0_RXD/MTR_START_BURNIN/WUU0_IN20
P4_13/ADC0_B5/ADC1_B5/TRIG_IN8/FC2_P1/USB1_ID/CT4_MAT1/FLEXIO0_D21/PLU_OUT1/CAN0_TXD
P4_14/CT4_MAT2/FLEXIO0_D22/PLU_OUT2
P4_15/ADC0_A1/CMP0_IN4P/TRIG_OUT4/USB1_VBUS_DIG/CT4_MAT3/FLEXIO0_D23/PLU_OUT3/WUU0_IN21
P4_16/ADC0_A6/FC2_P2/USB1_OTG_PWR/CT3_MAT0/FLEXIO0_D24/PLU_OUT4
P4_17/ADC0_B6/TRIG_IN9/FC2_P3/USB1_OTG_OC/CT3_MAT1/FLEXIO0_D25/PLU_OUT5
P4_18/CT3_MAT2/FLEXIO0_D26/PLU_OUT6
P4_19/ADC0_B1/CMP1_IN4P/TRIG_OUT5/CT3_MAT3/FLEXIO0_D27/PLU_OUT7
P4_20/ADC1_A6/TRIG_IN8/FC2_P4/CT2_MAT0/FLEXIO0_D28
P4_21/ADC1_B6/TRIG_IN9/FC2_P5/CT2_MAT1/FLEXIO0_D29
P4_22/CT2_MAT2/FLEXIO0_D30
P4_23/ADC0_A2/ADC0_B2/ADC1_B3/TRIG_OUT5/FC2_P6/CT2_MAT3/FLEXIO0_D31
MCXN547VDFT
U1F
P5_0/EXTAL32K/ADC1_B8/TRIG_IN10/LPTMR0_ALT2
P5_1/XTAL32K/ADC1_B9/TRIG_OUT6/LPTMR1_ALT2
P5_2/ADC1_B10/VBAT_WAKEUP_b/SPC_LPREQ/TAMPER0
P5_3/ADC1_B11/TRIG_IN11/RTC_CLKOUT/TAMPER1
P5_4/ADC1_B12/TRIG_OUT7/SPC_LPREQ/TAMPER2
P5_5/ADC1_B13/TRIG_IN10/LPTMR0_ALT2/TAMPER3
P5_6/ADC1_B14/TRIG_OUT6/LPTMR1_ALT2/TAMPER4
P5_7/ADC1_B15/TRIG_IN11/TAMPER5
P5_8/ADC1_B16/TRIG_OUT7/TAMPER6
P5_9/ADC1_B17/TAMPER7
MCXN547VDFT
U1G
RESET_B
USB0_DM/WUU0_IN28
USB0_DP/WUU0_IN29
USB1_DM
USB1_DP
USB1_ID
USB1_VBUS
ANA_0/ADC0_A0
ANA_1/ADC0_B0
ANA_4/ADC1_A0
ANA_5/ADC1_B0
ANA_6/ADC0_A3/ADC1_A3
ANA_14
ANA_18
ANA_22
MCXN547VDFT
VDD_P4
4.7K
4.7K
DNP
DNP
DNP
DNP
DNP
DNP
DNP
DNP
MCX-N5XX-EVK
MCU_PORT_4_5_A
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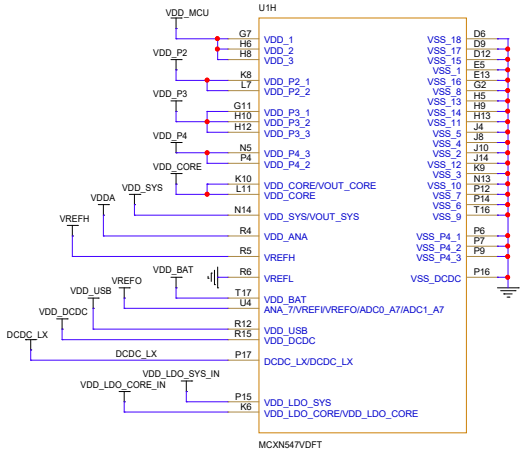
MCU POWER

No.	Power Config		On-chip Regulators											
		Module	DCDC		LDO Core		LDO Sys							
		Net	VDD_DCDC	DCDC_LX	VDD_LDO_CORE_IN	VDD_CORE	VDD_LDO_SYS_IN	VDD_SYS	VDD_MCU	VDD_P2	VDD_P3	VDD_P4	VDD_ANA	VBAT
		Jumper	JP27		JP18	JP35	JP36	/	JP14	JP47	JP15	JP45	JP39	JP37
1	MCU main power supply 3.3V, Port P3 1.8 V.	Setting	1-2		2-3	1-2	1-2	/	1-2	1-2	2-3	1-2	1-2	1-2
Power Efficient	The on-chip DCDC provides 1.1V VDD_CORE.	Connection	MCU_PWR	/	/	DCDC OUTPUT	MCU_PWR	SYS LDO OUT	MCU_PWR	P3V3	P1V8	P3V3	VDD_P4	MCU_PWR
Scheme(Default)	The on-chip LDO core is disabled	Status	Enable		Disable		Enable							
2	MCU main power supply 3.3V, Port P3 1.8 V.	Setting	3-4	5-6	1-2	Open	1-2	/	1-2	1-2	2-3	1-2	1-2	1-2
Low Cost	The on-chip DCDC is disabled.	Connection	GND	GND	MCU_PWR	CORE LDO OUT	MCU_PWR	SYS LDO OUT	MCU_PWR	P3V3	P1V8	P3V3	VDD_P4	MCU_PWR
Scheme	The on-chip LDO core is enabled to provide.	Status	Disable		Enable		Enable							

NOTE: MCU_PWR can be either 3.3V or 1.8V as determined by jumper JP38.

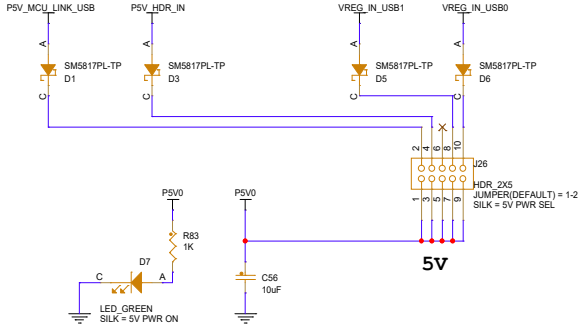


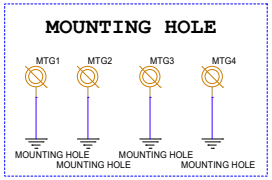


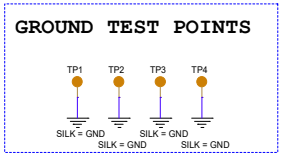


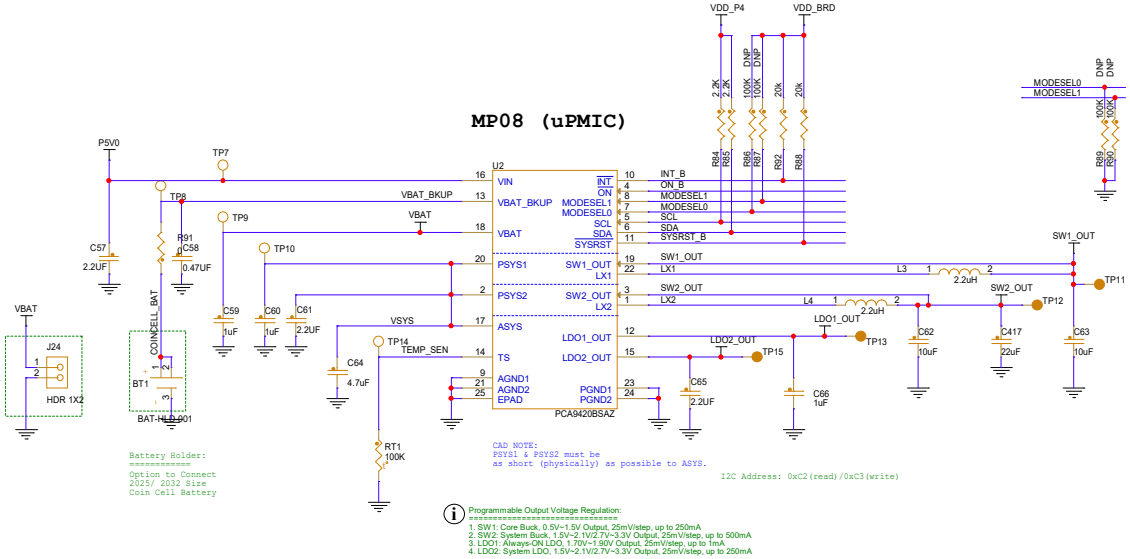
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Page Title:	MCU_POWER		
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PMIC POWER







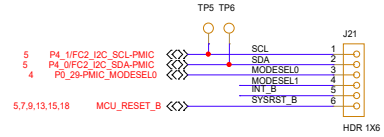


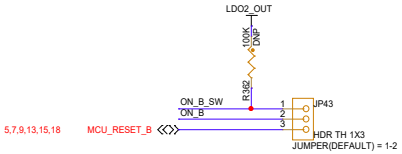
SW1 = 1.1V

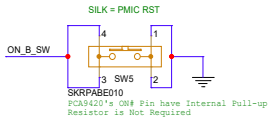
SW2 = 1.8V

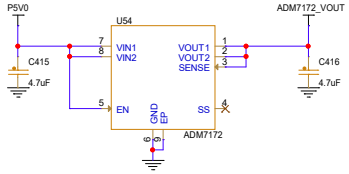
LDO1 = 1.8V

LDO2 = 3.3V

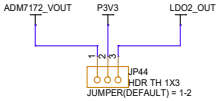




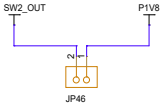




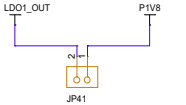
P3V3



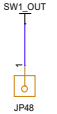
P1V8



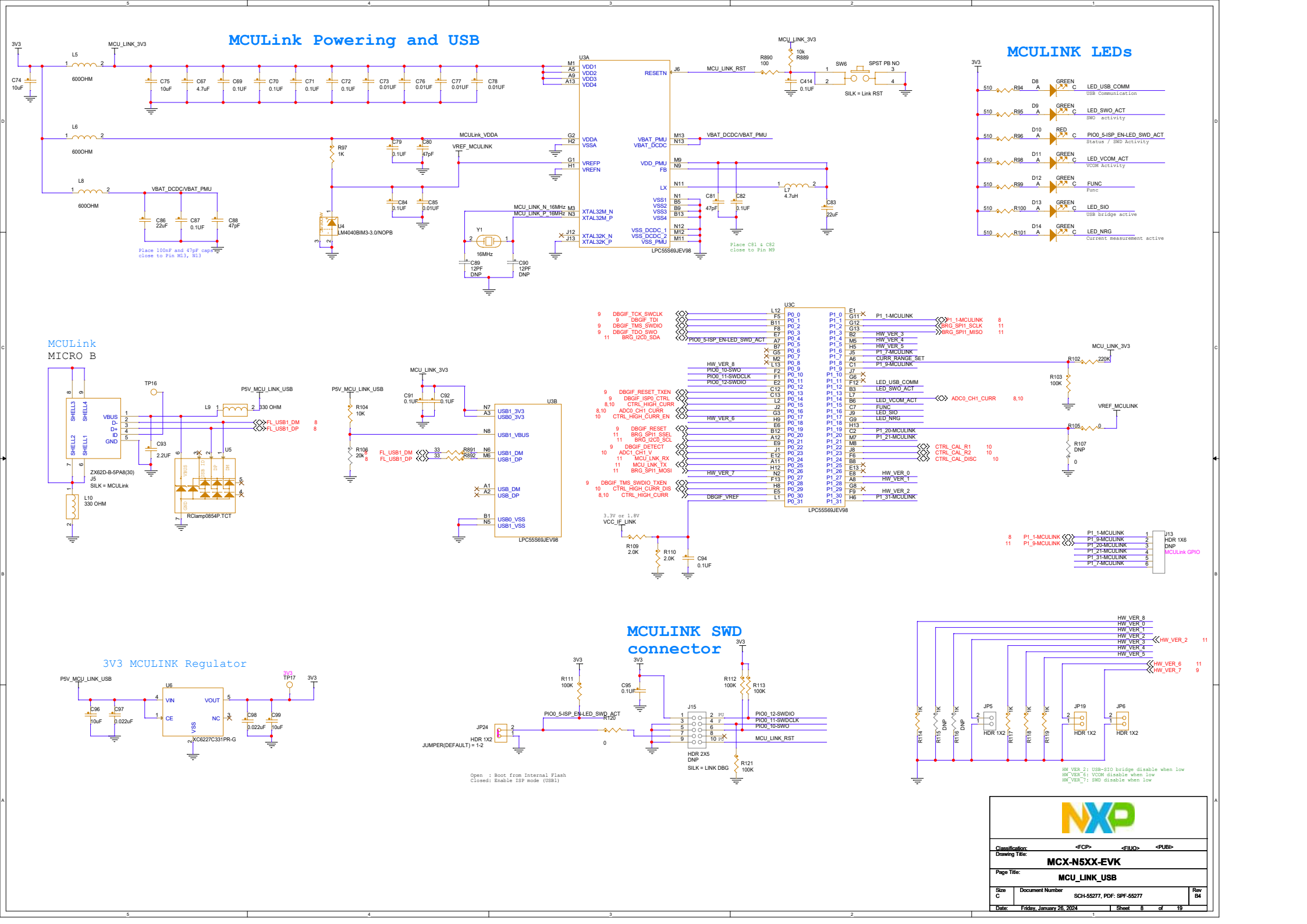
LDO1_OUT



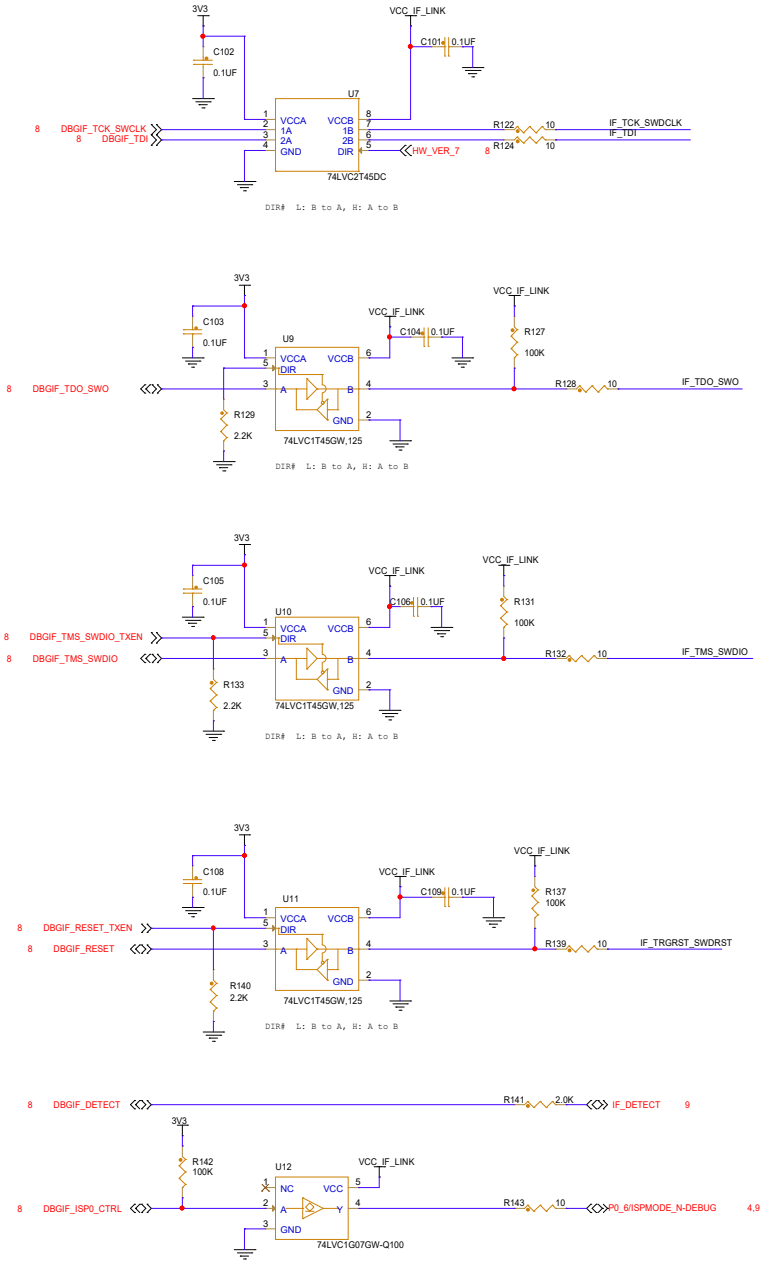
SW1_OUT

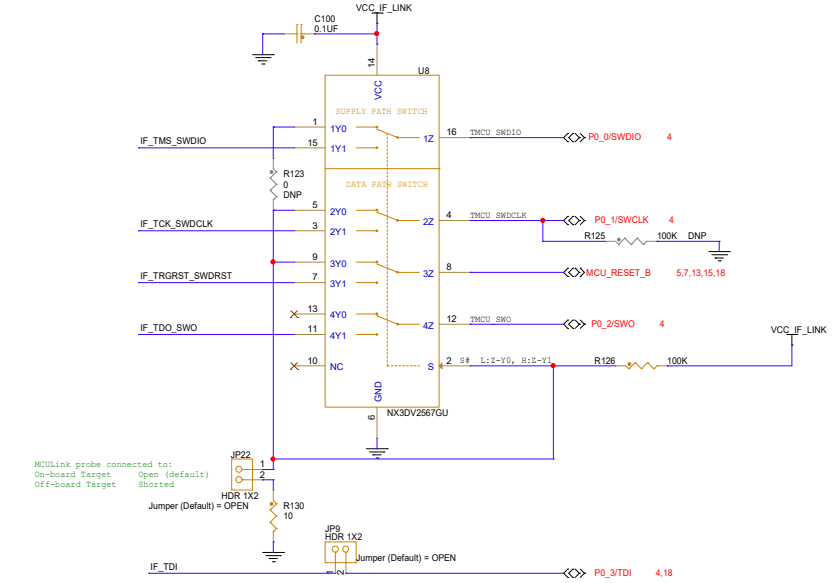


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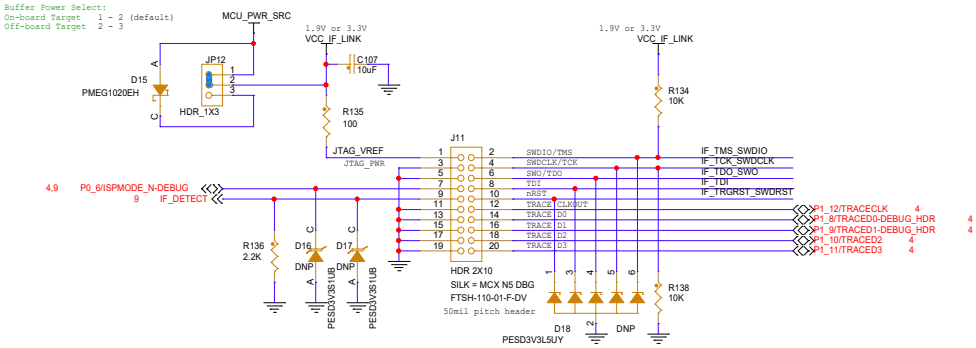

MCULink Powering and USB
MCULINK LEDs
MCULink
MICRO B
3V3 MCULINK Regulator
MCULINK SWD
connector
MCU_LINK_3V3
MCU_LINK_RST
MCULink_VDDA
VREF_MCULINK
VBAT_DCDC/VBAT_PMU
P5V_MCU_LINK_USB
LED_USB_COMM
LED_SWO_ACT
LED_VCOM_ACT
LED_SIO
LED_NRG
FUNC
LED_USB_BRIDGE_ACTIVE
LED_NRG
SILK = Link RST
SILK = MCULink
SILK = LINK DBG
Place 100nF and 47pF caps close to Pin M13, N13
Place C81 & C82 close to Pin M9
Open : Boot from Internal Flash
Closed: Enable ISP mode (USB1)
HW_VER_2: USB-SIO bridge disable when low
HW_VER_6: VCOM disable when low
HW_VER_7: SWD disable when low
MCULink GPIO
LPC55S69JEV98
RCLamp0854P.TCT
ZX62D-B-5PA8(30)
XC6227C331PR-G
LM4040BIM3-3.0/NOPB
Classification:
Drawing Title:
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MCULink Debug Interface



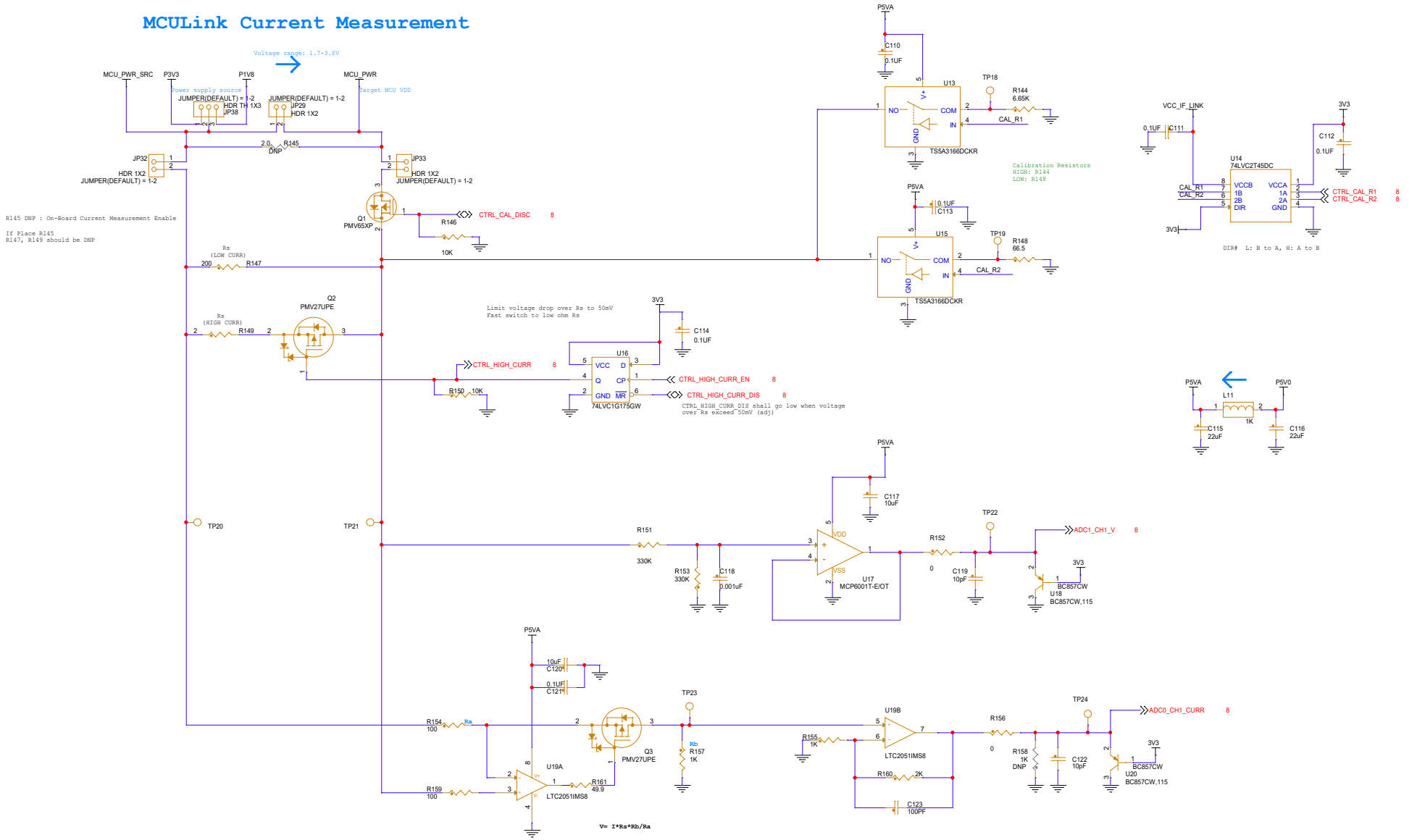


Debug Trace Connector



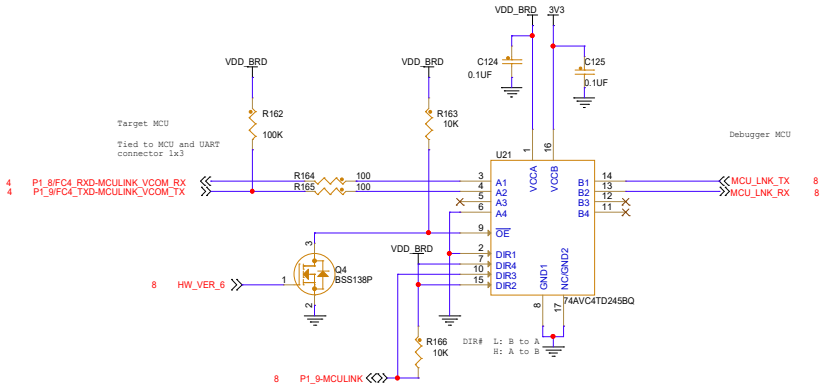
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Drawing Title:	MCX-N5XX-EVK		
Page Title:	MCU_LINK_DEBUG		
Size: C	Document Number: SCH-55277, PDF: SPF-55277		Rev: B4
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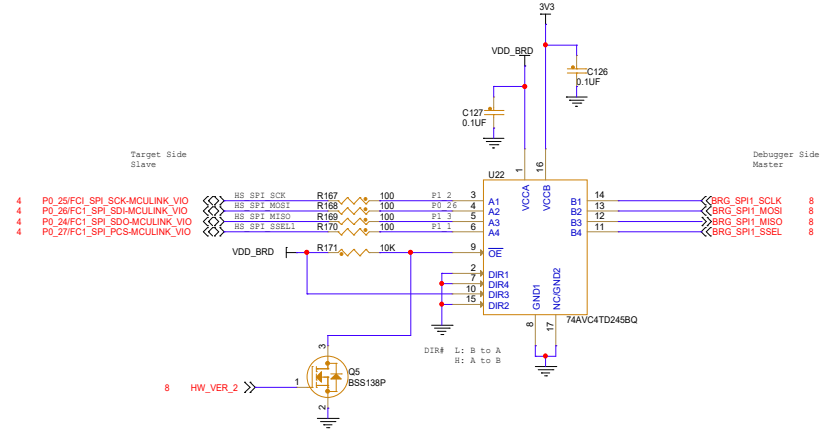
MCULink Current Measurement

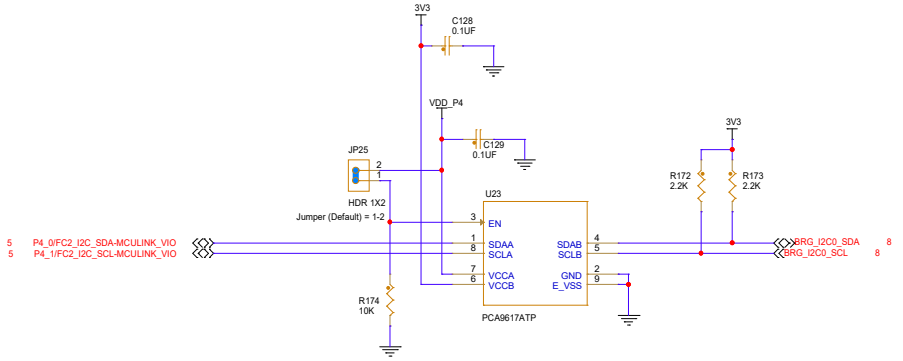


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Drawing Title:	MCX-N5XX-EVK		
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MCULink Voltage Translation Buffers



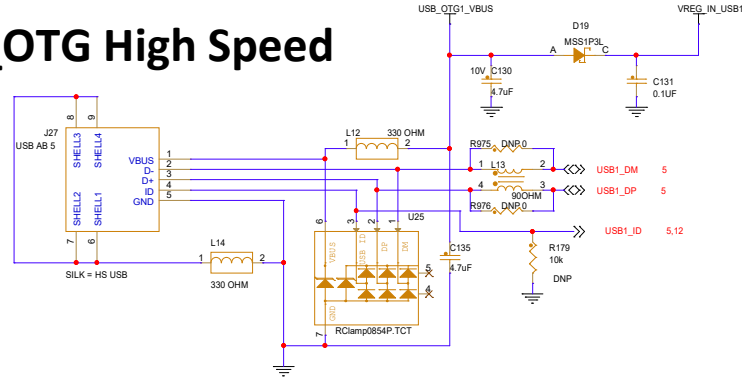




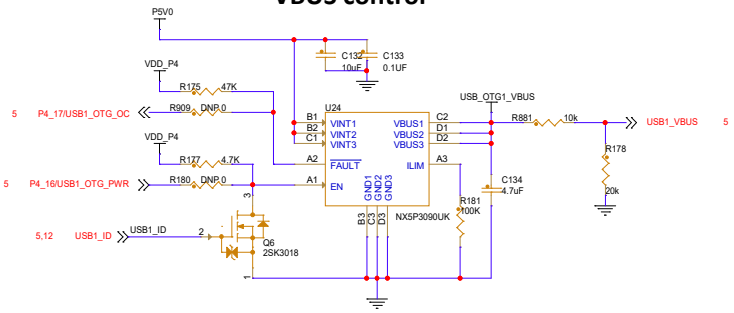
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Page Title:	MCU_LINK_VCOM_SIO		
Size: C	Document Number: SCH-55277, PDF: SPF-55277		Rev: B4
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USB

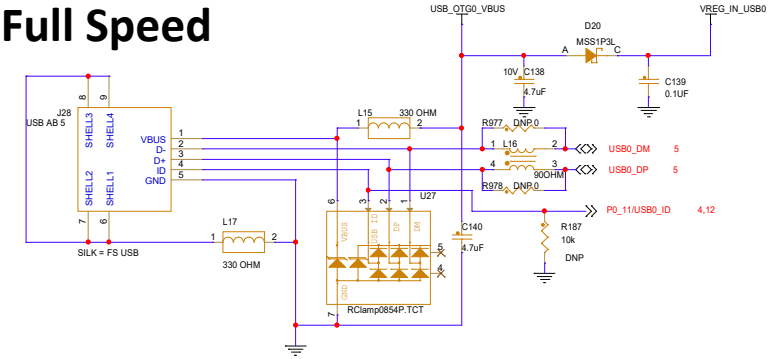
USB1_OTG High Speed



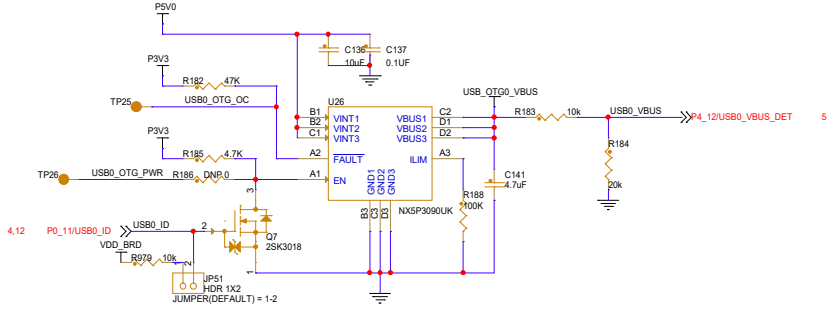
VBUS control



USB0 Full Speed



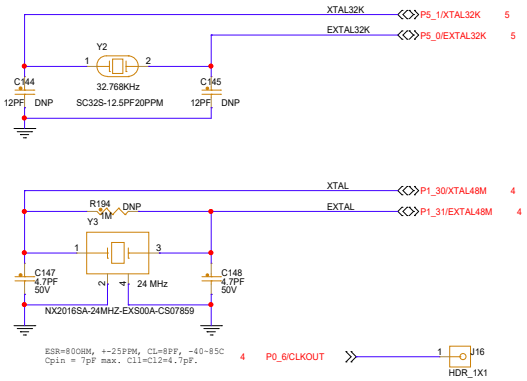
VBUS control



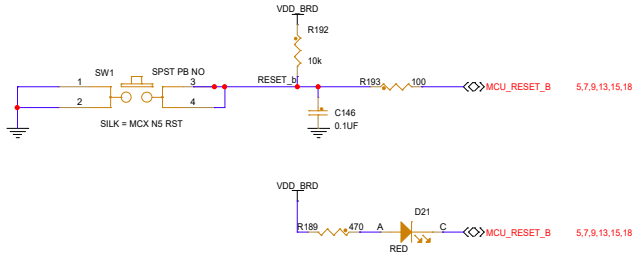
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Page Title:	USB		
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CLOCK & RESET & HMI

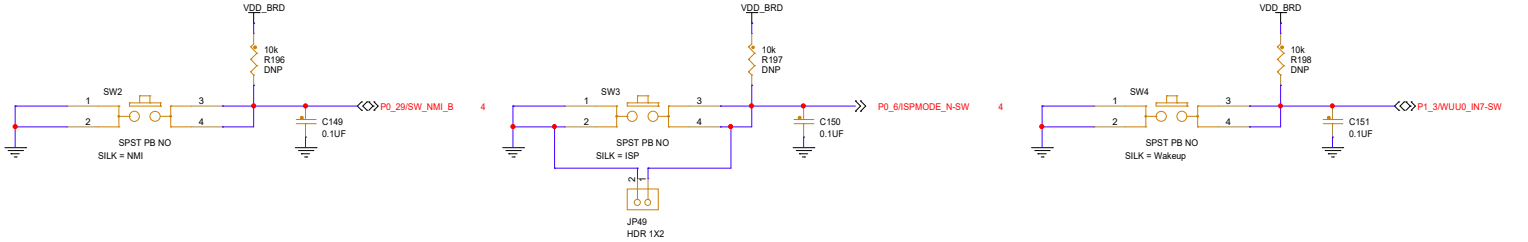
Clock



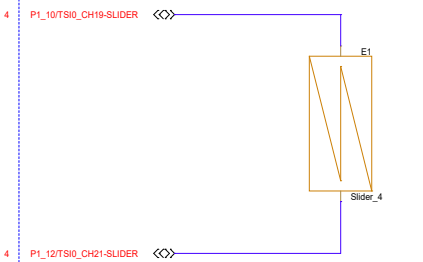
MCU Reset



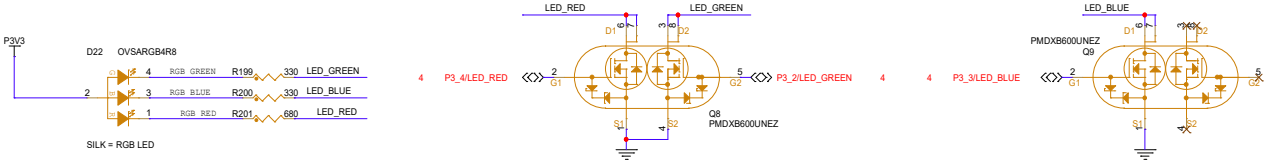
Push Buttons



Touch Slider

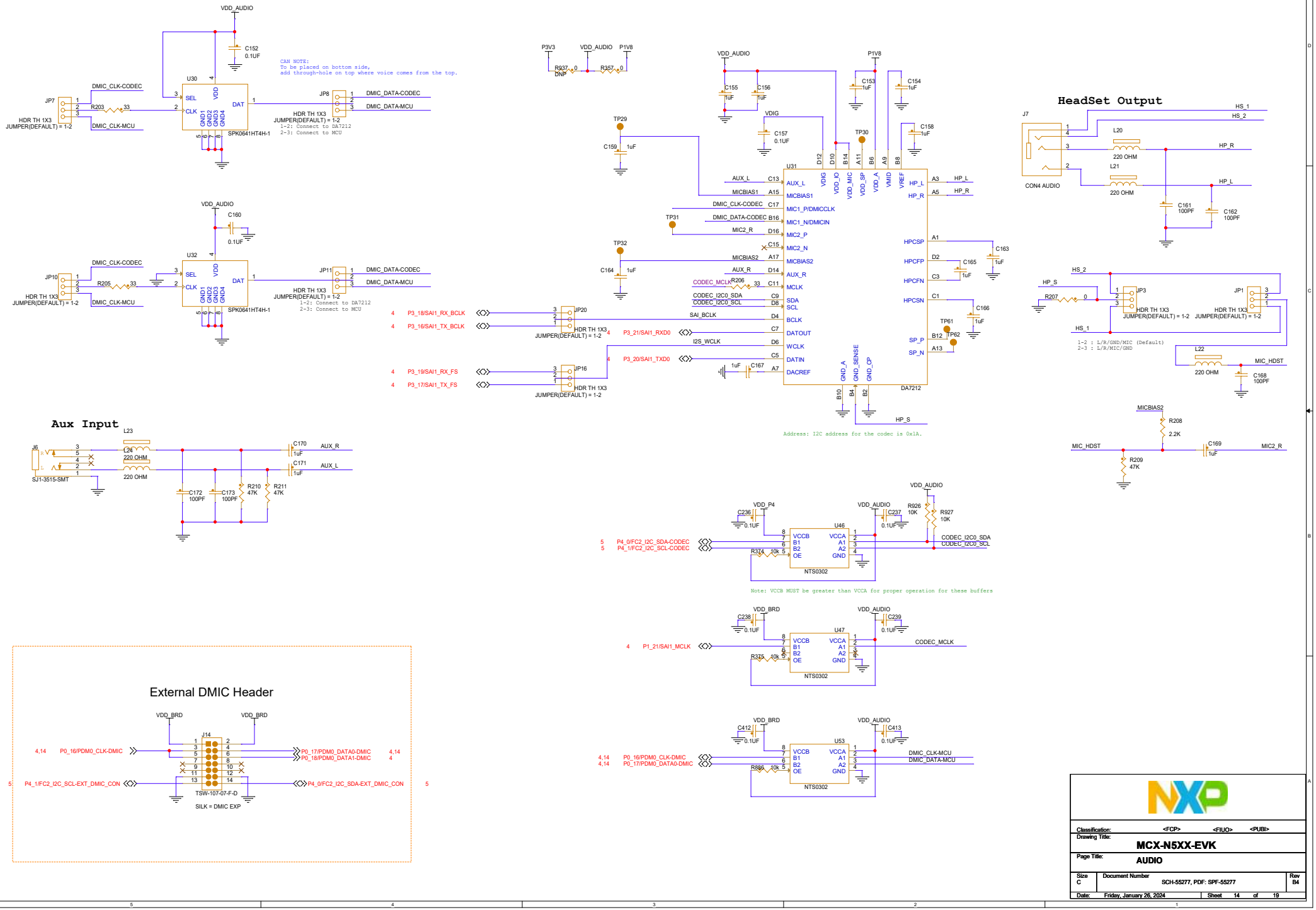


RGB LED



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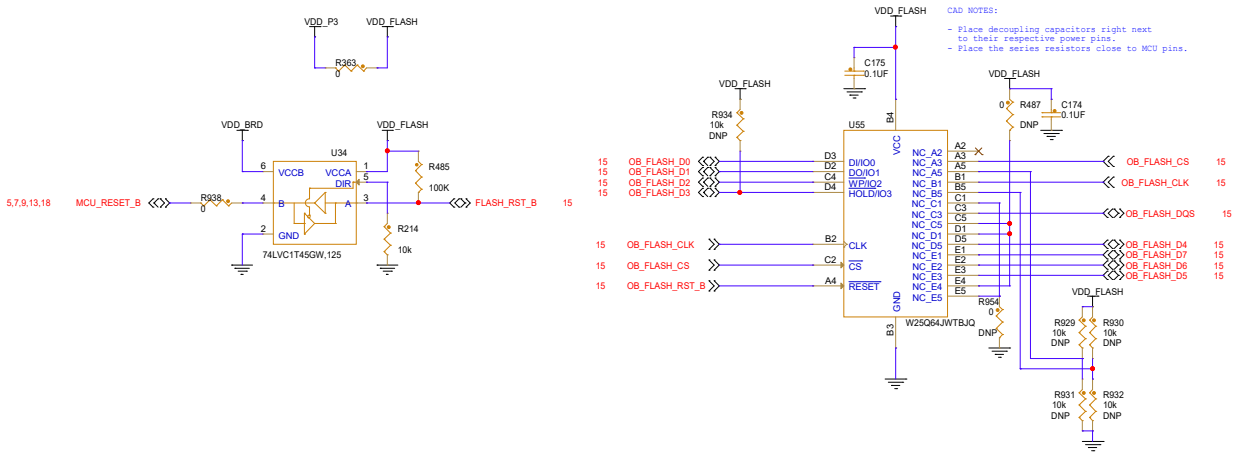
AUDIO CODEC, DIGITAL MIC AND CONNECTORS



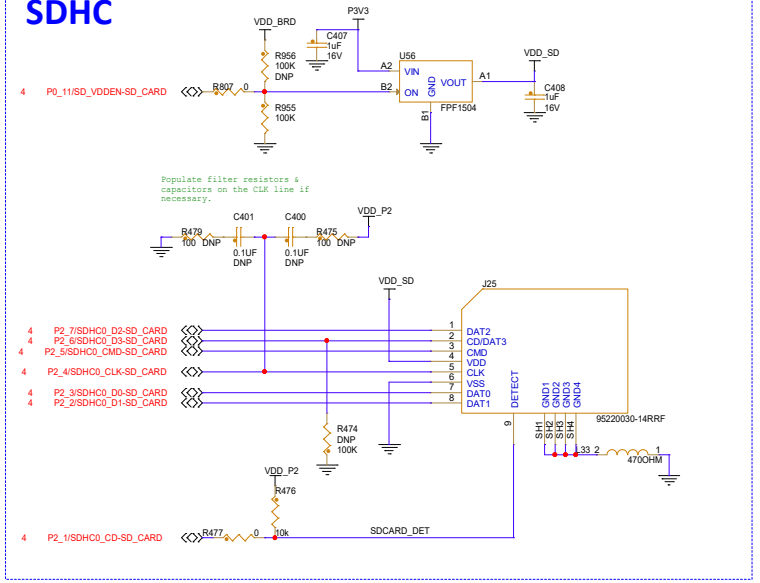
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Page Title:	AUDIO		
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W25Q64JWTBJQ only
supports 1.8V operation.
VDD_P3 must be 1.8V.

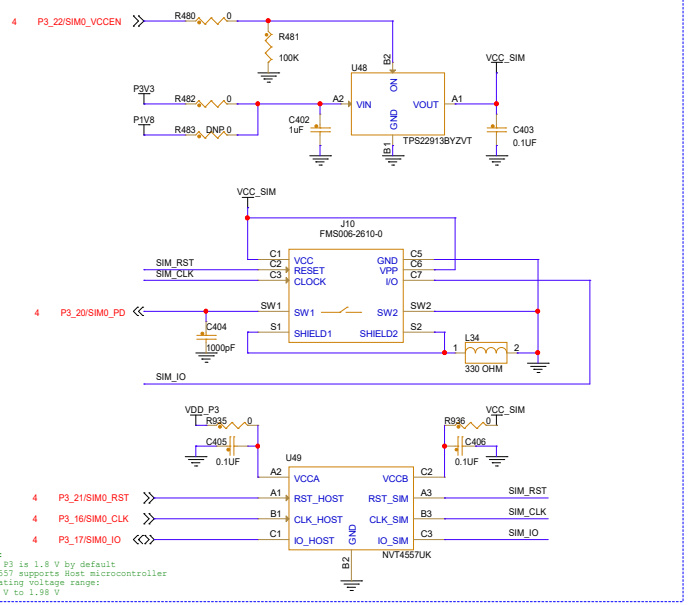
FlexSPI Memory

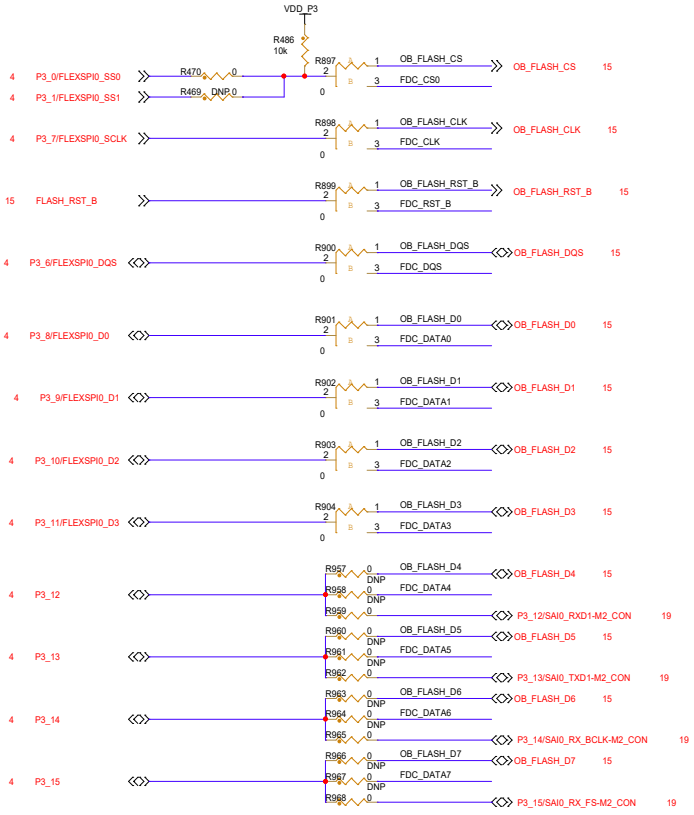


SDHC



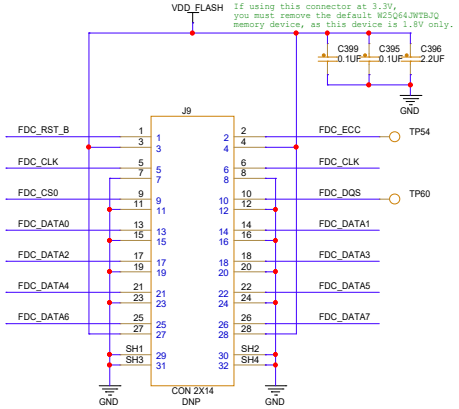
EMVSIM





CAD note:
Resistors for P3_12, P3_13, P3_14, and P3_15
need to be placed and routed to minimize stubs.

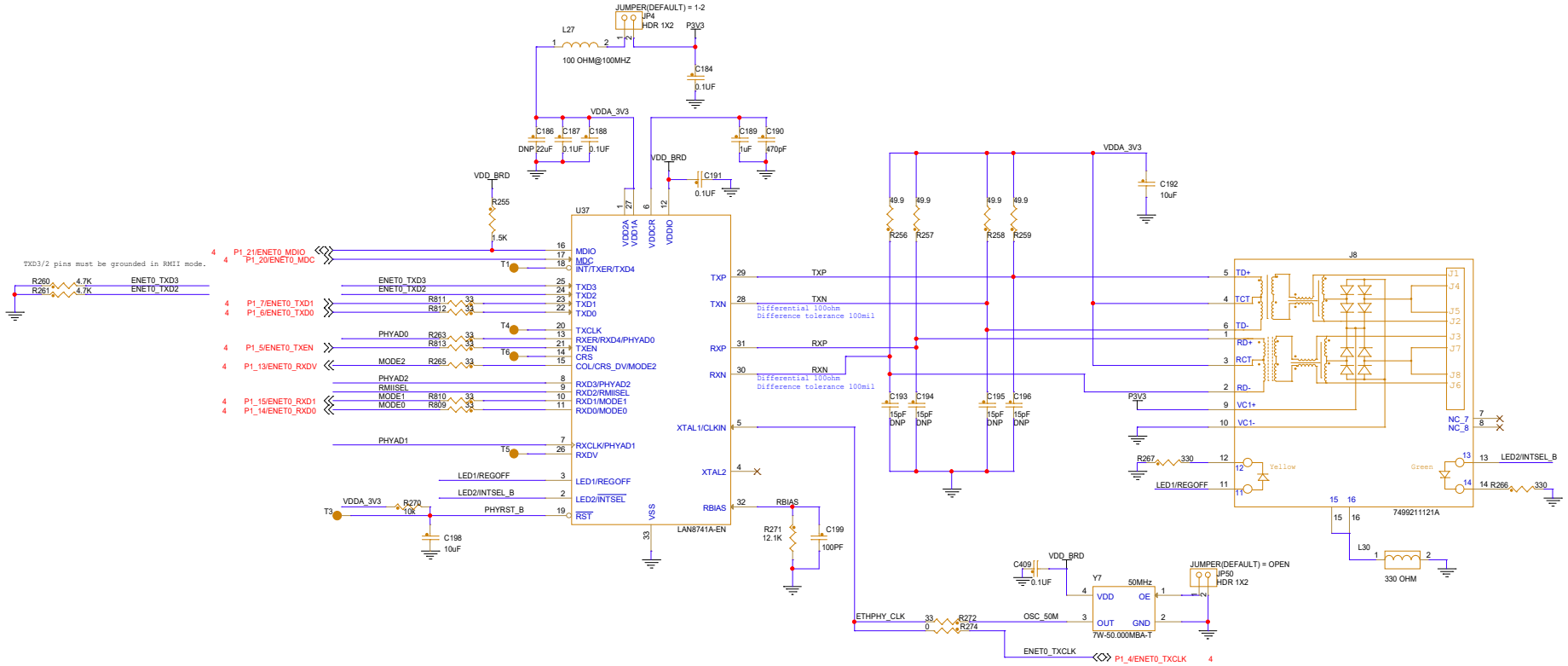
FLASH DAUGHTER CARD



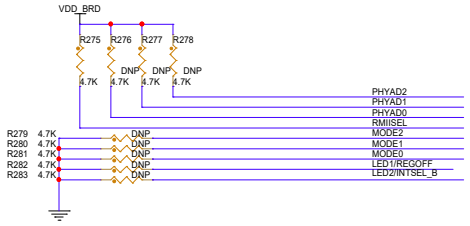
PINOUT INFO
==========
3/ 4: Flash core voltage
27/28: Flash IO voltage
(connected on this board)

Classification:	<FCP>	<FIUO>	<PUBI>
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ETHERNET (RMII)



PHY Strap-In Options



RJ45 Connector

Config Pin (Internal State)	Description	Jumper Setting
MODE[0:2] (PU)	Mode Config 111 => All Capable, auto-neg enabled (Default)	Open (Int PU)
PHYAD[0:2] (PD)	Device Address 000 => Default	Open (Int PU)
RMIISEL (PD)	1 => RMII (Default) 0=> MII	Closed (Ext PU)
REGOFF (PD)	1 => Internal Regulator OFF 0=> Internal Regulator ON (Default)	Open (Int PU)
INTSEL_B (PU)	Floating 1 => nINT function (Default) 0=> TXD4/TXER function	Open (Int PU)

Classification: <FCP> <FIUO> <PUBI>

Drawing Title: MCX-N5XX-EVK

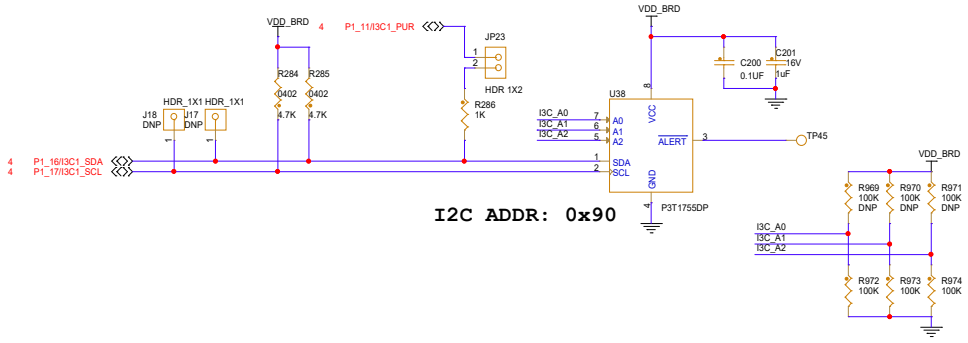
Page Title: ETHERNET

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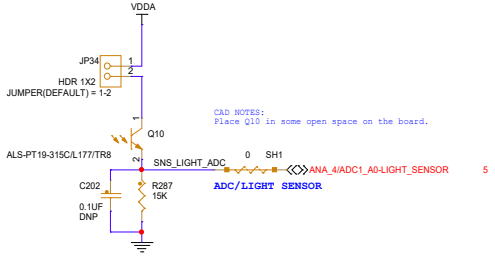
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SENSOR & CAN

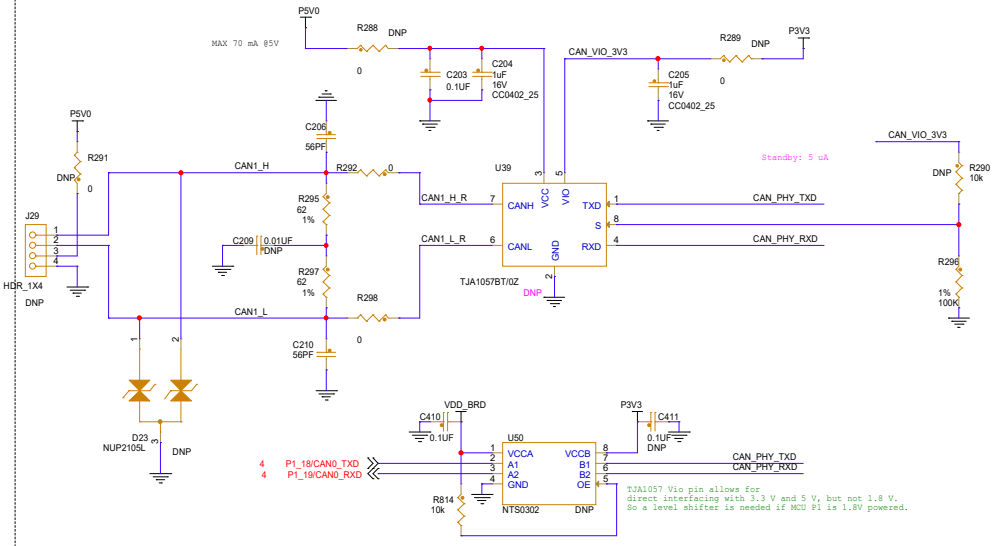
I3C Sensor



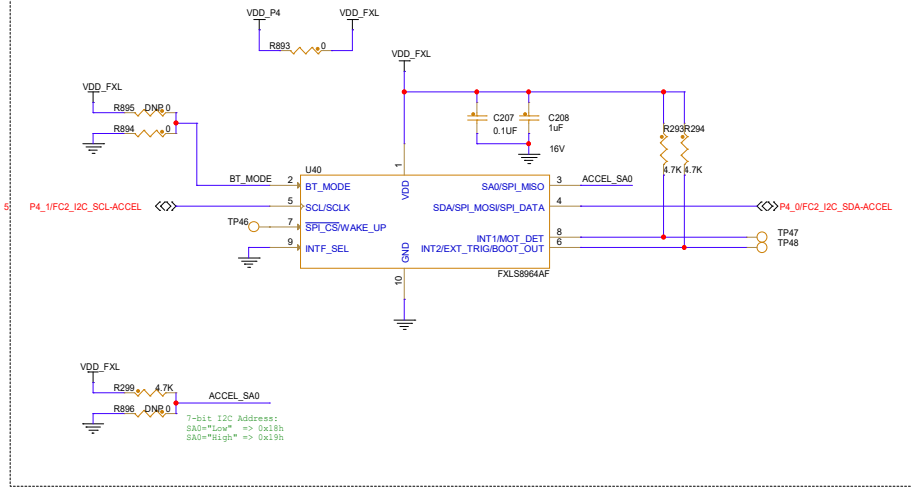
Visible Light Sensor



FlexCAN

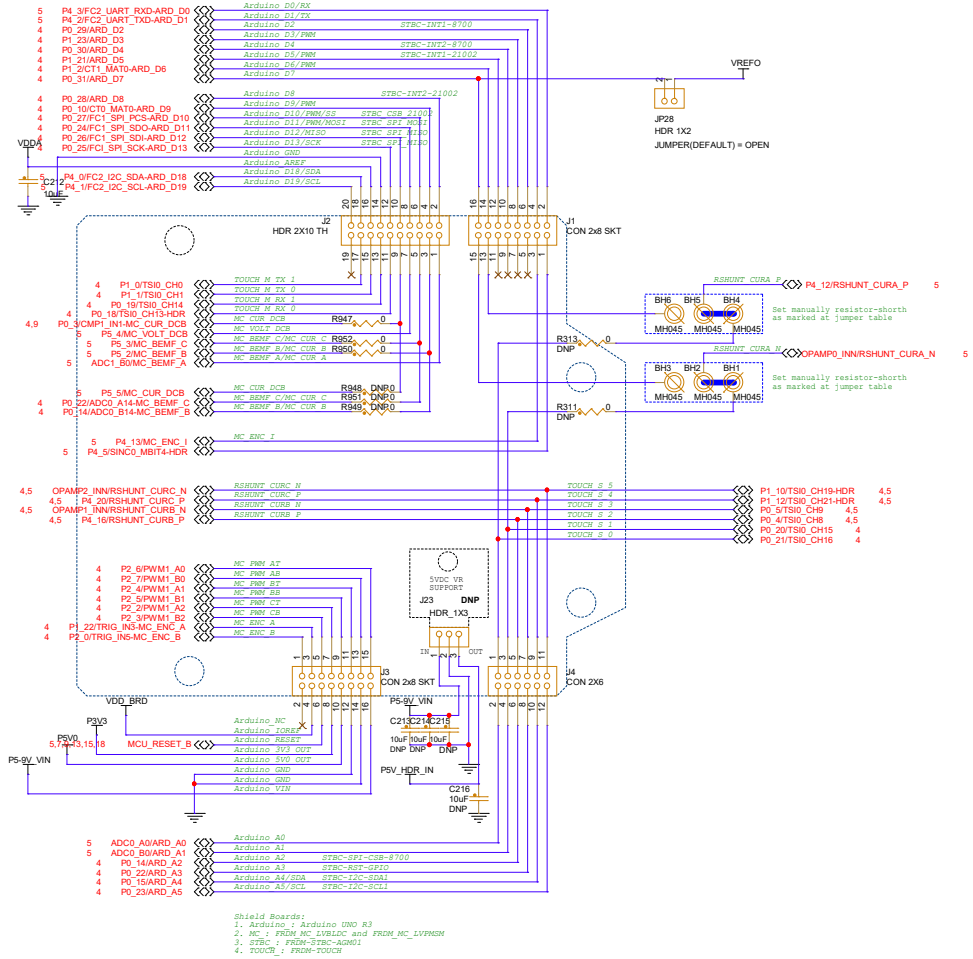


3-Axis Accelerometer

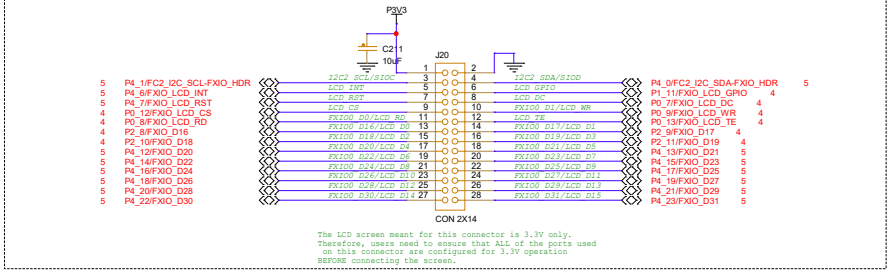


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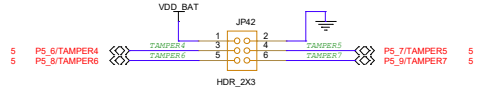
ARDUINO SHIELD COMPATIBLE HEADERS



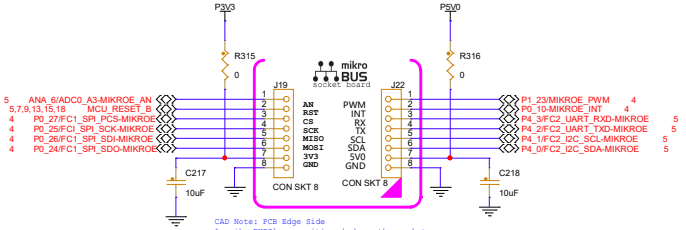
FLEXIO HEADER



TAMPER HEADER



mikroBUS

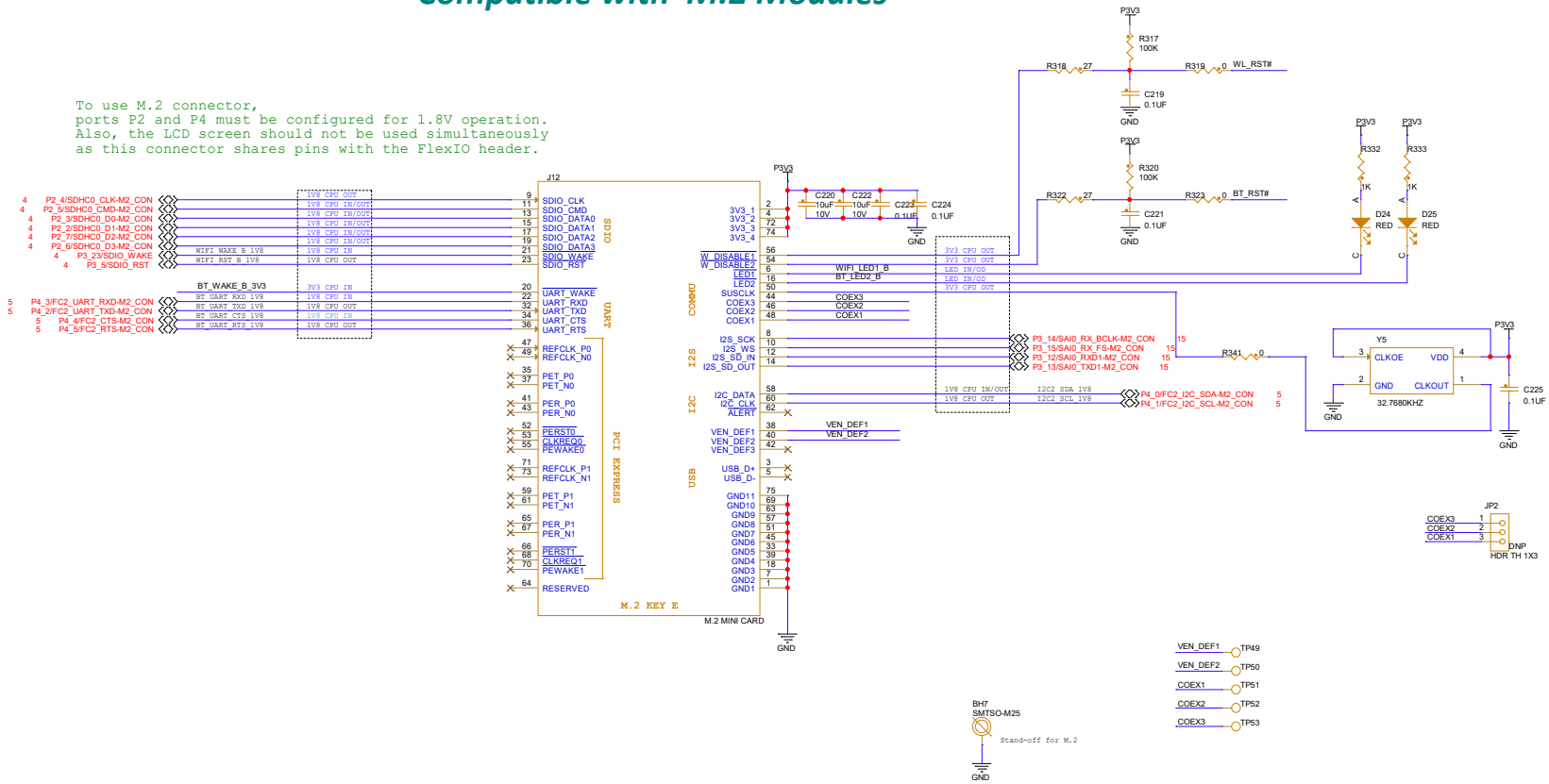


CAD Note: PCB Edge Side
1. mikroBUS?logo positioned above the socket
2. Pin names corresponding to the routing of the mainboard
3. A line that encloses the socket on all sides except the top.
4. A diagonal notch below the right-hand side pinout (serves as a guideline for add-on board insertion)

Classification:	<FCP>	<FIUO>	<PUBI>
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M.2 CONNETOR

Compatible with M.2 Modules



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